MIL-C-52853B
23 July 1980
SUPERSEDING
MIL-C-52853A(ME)
10 March 1977

MILITARY SPECIFICATION

CONTROL UNIT, ALARM SET:

C-9412/FSS-9(V)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the intrusion alarm set control unit. It is used to process signals from intrusion detection sensors and to obtain a continuous status of an area which has been secured.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781 - Strapping, Steel, and Seals.
PPP-B-566 - Boxes, Folding, Paperboard.

FSC 6350

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Setup.

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-C-45662	- Calibration System Requirements.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment, Subsystem and System.
MIL-STD-781	- Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution.
MIL-STD-810	- Environmental Test Methods.

DRAWINGS

ME

TA13220E4200	- Control Unit, Alarm Set C-9412/FSS-9(V).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

TA13220E4202	- Power Supply Control Unit.
TA13220E4213	- Printed Wiring Board Assembly, Tamper/Alarm.
TA13220E4214	- Printed Wiring Board Assembly, Voltage Monitor.
TA13220E4215	- Printed Wiring Board Assembly, Audible Alarm.
TA13220E4216	- Printed Wiring Board Assembly, Instantaneous Alarm.
TA13220E4217	- Printed Wiring Board Assembly, Latched Alarm.
TA13220E4218	- Printed Wiring Board Assembly, Alarm I.D.
TA13220E4219	- Printed Wiring Board Assembly, L.E.D.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The alarm set control unit, hereinafter called "control unit", shall be in accordance with Top Assembly TA13220E4200 and as specified herein.

3.1.1 Major components. The control unit consists of the following major components:

(a) Enclosure assembly.
(b) Power supply assembly TA13220E4202.
(c) Battery 24 Vdc, 18 Amp-hr C13220E4208.
(d) Status processor assembly.
(e) Printed wiring board assembly, tamper/alarm TA13220E4213.
(f) Printed wiring board assembly, voltage monitor TA13220E4214.
(g) Printed wiring board assembly, audible alarm TA13220E4215.
(h) Printed wiring board assembly, instantaneous alarm TA13220E4216.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:10Z
Check the source to verify that this is the current version before use.

(i) Printed wiring board assembly, latched alarm TA13220E4217.
(j) Printed wiring board assembly, alarm I.D. TA13220E4218.
(k) Printed wiring board assembly, L.E.D. TA13220E4219.

When specified (see 6.2), any of the major components listed above shall be furnished separately.

3.1.2 Function. The control unit is the central control element of J-SIIDS. Its primary function is to process signals from intrusion detection sensors and to obtain a continuous status of the controlled zone. This information is applied to output (data link) terminals for transmittal to periphery monitor equipment. The control unit also supplies primary and emergency power to its associated sensors; provides "secure", "access", and "test" modes of system operation within the controlled zone; and maintains control zone subsystem security by monitoring sensor and audible alarm signal lines, and antitamper control of its own enclosure.

3.1.3 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the control unit. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The contractor shall furnish one or more control units and major components furnished separately, as specified (see 6.2), for examination and testing within the time frame specified (see 6.2), to prove prior to starting production that his production methods will produce control units and major components furnished separately that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4; and unless otherwise specified herein, all examinations and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

Government will conduct any or all of the preproduction examination and tests, as specified (see 6.2).

3.3 Initial production. When specified (see 6.2), the contractor shall furnish to the Government one or more control units for inspection as specified in 4.4.

3.4 Material. Material shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.9).

3.5 Environmental conditions. The control unit and major components furnished separately, shall withstand individual environmental conditions specified herein, or combinations thereof, without damage (see 6.4.3), performance degradation, or malfunction when tested as specified in 4.6.2.5.

3.5.1 Operational.

(a) Temperature range: -20° F to +150° F.
(b) Humidity: up to 95 percent relative.

3.5.2 Nonoperational and storage.

(a) Temperature range: -30° F to +165° F.
(b) Humidity: up to 95 percent relative.
(c) Shock:
    (1) 20 times the acceleration due to gravity (g's) for 11 milliseconds (ms). Transportation conditions (without battery).
    (2) Bench handling. Equipment not operating (with battery).
(d) Vibration: Transportation conditions (without battery).

3.6 Detailed technical requirements.

3.6.1 Detailed design. The control unit shall not false alarm (see 6.4.1) under any of the operating test conditions specified herein.

3.6.1.1 Electrical interfaces. The electrical signal interface between the control unit, intrusion sensors, latching alarm switch, audible alarm and the data link shall be standardized for complete interchangeability of these components of J-SIIDS. To facilitate this standardized interface, the terminal characteristics on the inter-

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

connecting transmission lines shall be as specified in 3.6.1.1.1 and 3.6.1.1.2 for all standard alarm signals (see 6.4.2). The electrical signal interface between the control unit and the audible alarm (when used as part of J-SIIDS) differs from the standard alarm signals defined above. These interfaces are specified in 3.6.1.5.1.3 and 3.6.1.5.2.

3.6.1.1.1 Receiving end. The input circuits in the control unit shall provide a direct current to the interconnecting lines which shall not exceed 10 milliamperes (mA) under short circuit conditions, nor develop over 30 volts direct current (Vdc) under open circuit conditions. The control unit shall recognize and react to the following input alarm conditions when the signals are maintained for a period of not less than 100 ms:

Input Alarm Conditions

No alarm: 0 to 20 kilohms (k ohms).
Alarm: more than 100k ohms.

3.6.1.1.2 Sending end. The output circuits in the control unit shall function properly at line currents up to 10 mA and voltages up to 30 Vdc. The control unit shall continuously present the following output alarm conditions to the interconnecting lines:

Output Alarm Conditions

No alarm: 0 to 2k ohms.
Alarm: more than 100k ohms.

3.6.1.2 Enclosure assembly. The enclosure assembly shall be in accordance with the applicable drawings. Installation of switches, terminal boards, and interconnecting wiring shall be in accordance with Top Assembly TA13220E4200. All interconnecting wiring between terminal boards shall be installed and secured to facilitate removal and replacement of modular assemblies and installation of interconnecting wiring.

3.6.1.2.1 Tamper protection. The enclosure door and tamper switch shall be assembled such that the switch contacts open with not less than 0.125-inch nor more than 0.250-inch of door movement and before the door flanges are disengaged.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:10Z
Check the source to verify that this is the current version before use.

3.6.1.2.2 Terminal board connections. Terminal boards and terminals shall be marked for identification as follows:

(Intrusion Sensors)

Terminal Boards A- through E-	Function
1	Intrusion alarm signal
2	Signal common
3	Tamper alarm signal
4	Sensor power (20 Vdc regulated)
5	Sensor power return
6	Ground
7	Access command signal (27.5 Vdc unregulated)

(Latching Alarm Switch)

Terminal Board F-	Function
1	Duress alarm signal
2	Signal common
3	Tamper alarm signal
4	Sensor power (20 Vdc regulated)
5	Sensor power return
6	Ground
7	Spare

(Audible Alarm)

Terminal Board L-1-	Function
1	High, ac
2	Neutral, ac
3	Ground

(Audible Alarm)

Terminal Board L-2-	Function
1	Alarm signal
2	Signal common
3	Tamper alarm signal
4	5 Vdc

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

(Data Link)

Terminal Board S-	Function
1-I	Instantaneous alarm signal
1-IR	Instantaneous alarm signal return
1-D	Latched alarm signal
1-DR	Latched alarm signal return
2	Access-secure signal
2R	Access-secure signal return
3	Power, ac, on/off signal
3R	Power, ac, signal return
4	Ground
5	20 Vdc regulated
6	Regulated dc return
7	Resynchronization (Resync) Initiation
8	Telephone dialer tamper signal
9	Telephone dialer tamper return

3.6.1.3 Power supply assembly. The power supply assembly shall be in accordance with Top Assembly TA13220E4202 and as specified herein.

3.6.1.3.1 Power supply. The power supply assembly shall operate on primary power of 110 to 125 Volts alternating current (Vac), 48 to 62 Hertz (Hz) and provide one ac and four dc outputs. In event of failure of ac power, the power supply assembly shall provide three dc outputs while operating on standby dc power from one 24 Vdc, 18 ampere-hour lead-calcium battery. The five outputs of the power supply assembly shall have the following characteristics:

Output number	Output voltage	Line regulation	Load regulation	Ripple content
1	20 ±1 Vdc at 0.6 amperes (A)	±0.5%	±1.0% no load to rated load	300 millivolt (mV) peak-to-peak at 0.6 A.
2 1/	27 ±0.5 Vdc at 4.0 A	±1.0%	Not applicable	Not applicable

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:10Z
Check the source to verify that this is the current version before use.

Output number	Output voltage	Line regulation	Load regulation	Ripple content
3	5 ±0.25 Vdc at 50 mA	±1.0%	±1.0% no load to rated load	200 mV peak-to-peak at 50 mA.
4	6.3 Vac at 0.2 A	±10%	±5% no load to rated load	Not applicable
5 2/	27 ±0.5 Vdc at 0.25 A	±1.0%	±1.0% no load to rated load	Not applicable

1/ This output is for fast charging of the 24-volt lead-calcium standby battery. A fast charging bypass circuit shall provide 26 to 30 Vdc unregulated at 4 A directly at Output 2.

2/ This output is for trickle charging battery.

3.6.1.3.2 Power supply, ac presence. The power supply shall sense the presence of the primary ac power input. Interruption of the primary ac power input shall cause a high impedance greater than 100K ohms between terminals TB3-1 and TB3-2 after a 3-second interval. Resumption of primary ac power input shall immediately return a low impedance condition of less than 2k ohms between terminals TB3-1 and TB3-2.

3.6.1.3.3 Battery charger. Output 2, 26 to 30 Vdc unregulated, shall be suitable for fast charging the lead-calcium standby battery and shall not overcharge the battery when operated on ac voltage specified in 3.6.1.3.1. The charger shall bring the battery from 10 percent of full charge to 90 percent of full charge in 24 hours at any operating temperature.

3.6.1.4 Standby power supply. Standby power shall be supplied from a 24-volt lead-calcium battery. The battery shall be in accordance with Drawing C13220E4208. The 24-volt battery shall have sufficient capacity to maintain the control unit and its associated connected units in operation for 24 hours at any operating temperature above 0° F with the voltage maintained at or above 21 volts throughout the interval. Batteries shall be fully charged prior to shipment to the specified destination in the contract. Date of battery charging shall be marked on the battery box (see 5.1.1.1.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

3.6.1.5 Status processor assembly. The status processor assembly shall be in accordance with the applicable drawings. It shall receive and process all alarm signals in addition to the ac power status and the regulated dc voltage status, and present the control zone and control unit status.

3.6.1.5.1 Input signals.

3.6.1.5.1.1 Duress alarm. The status processor shall accept standard input alarm conditions from sensor input terminal F-1.

3.6.1.5.1.2 Intrusion alarms. The status processor shall accept standard input alarm conditions from sensor input terminals A-1, B-1, C-1, D-1, and E-1.

3.6.1.5.1.3 Tamper alarms. The status processor shall accept standard input alarm conditions from sensor input terminals A-3, B-3, C-3, D-3, E-3, F-3, and S-8; from internal tamper switch S4; and from the ac power monitor in the control unit power supply. The status processor shall recognize a no-alarm condition when the control unit power supply regulated 20-volt supply is between 19 and 21 volts and an alarm condition when the regulated supply is over or under the specified voltage. The regulated dc power status shall be processed as tamper alarms. The status processor shall also accept tamper alarm inputs from audible alarm input terminal L-2-3. The status processor shall recognize and react to a no-alarm condition when the voltage across terminals L-2-3 and L-2-2 is between 4 and 5 Vdc and to an alarm condition when the voltage across these terminals is less than 1 Vdc.

3.6.1.5.2 Output signals. After processing alarms as specified herein, the status processor shall provide standard output alarm conditions to the following output terminals:

Terminals	Condition (Impedance State)
S-1-I to S-1-IR	Instantaneous alarm (high), No-alarm (low)
S-1-D to S-1-DR	Latched alarm (high), No-alarm (low)
S-2 to S-2R	Secure (high), Test/reset or access (low)
S-3 to S-3R	Power, ac, off (high); On (low)

Alarm outputs shall also be provided between output terminals L-2-1 and L-2-2. The internal impedance between these terminals shall be 1k ohm ±10 percent. A voltage of 4 to 5 Vdc at 1.7 mA shall be applied between these terminals for a no-alarm condition, and reduced to less than 1 volt for an alarm condition.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

3.6.1.5.3 Signal processing.

3.6.1.5.3.1 Instantaneous alarm output. All alarm conditions presented between output terminal S-1-I and its return shall appear for 10 ±2 seconds after initiation. If an alarm input is present at the control unit inputs or is still present from the internal tamper circuit or the fail-safe circuits after this time period, the alarm signal between S-1-I and its return shall continue until the alarm input disappears.

3.6.1.5.3.2 Latched alarm output. A dual-mode of operation shall be provided for the latched alarm output presented between output terminal S-1-D and its return. The mode of operation shall be selected by the latch/nonlatch switch located within the status processor. In the latch mode of operation, all alarm conditions presented between output terminal S-1-D and its return and between L-2-1 and L-2-2 shall be maintained from initiation until the control unit is reset by rotating the operating mode switch to the TEST/RESET position. In the nonlatch mode of operation, all alarm conditions presented between output terminal S-1-D and its return shall be maintained from initiation for a period of 10 ±2 seconds. The alarm output presented between L-2-1 and L-2-2 shall not be affected, but shall be maintained until reset by rotating the operating mode switch to the TEST/RESET position.

3.6.1.5.3.3 Intrusion alarm time delay functions. Intrusion alarms initiated within an adjustable time delay period (exit delay) of 10 ±2 to 90 ±10 seconds after the operating mode switch is rotated to the SECURE position shall be inhibited from appearing between terminals L-2-1 and L-2-2 and between terminal S-1-D and its return. These intrusion alarms shall, however, appear instantaneously between S-1-I and its return. Tamper alarms appearing within this exit delay period shall not be so inhibited but shall appear instantaneously at all alarm output terminals. Duress alarms shall also not be inhibited between output terminal S-1-I and its return and between output terminal S-1-D and its return. Intrusion alarms initiated after the expiration of this exit delay shall appear between S-1-D and its return and between L-2-1 and L-2-2 after the expiration of the entrance time delay, adjustable between 10 ±2 to 90 ±10 seconds unless inhibited by the reset function when the operating mode switch is moved to the TEST/RESET position prior to the expiration of the entrance time delay. Tamper and duress alarms, when presented between terminal S-1-I and its return, shall appear instantaneously and not be delayed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

3.6.1.5.4 Alarm identification. Turning the operating mode switch to SECURE shall extinguish all alarm identification light emitting diodes (LED's). After the exit time delay expires, the circuit shall be armed to activate the LED associated with the first intrusion or duress alarm input to occur. The first intrusion or duress alarm input to occur after the circuit is armed shall light the LED associated with its alarm input circuit. If the first alarm input is followed within 1 second or less by additional intrusion or duress alarms to different alarm input circuits, the LED's associated with these alarm inputs may also light; however, after a maximum of 1 second has passed from the time of the first alarm input, this circuit card shall be de-armed. Intrusion or duress alarms occurring after the circuit is de-armed shall not cause any LED to light; however, the LED's which were activated during the armed period shall be held on until the operating mode switch is next turned from the ACCESS or TEST/RESET position to the SECURE position. Removal of this circuit card shall not affect the ability of the control unit to meet the other performance requirements of this specification.

3.6.1.6 Operating mode control. The operating mode switch shall be a three-position, key-operated, double-action, high security type switch. Two keys shall be furnished for the lock. The key shall be removable only in the SECURE position. The switch shall be capable of 50,000 switching operations opening a 30-volt, 100 mA dc circuit (resistive load), and shall be sealed against dirt and dust. The control functions in each of the three positions shall be as follows.

3.6.1.6.1 ACCESS. Intrusion alarms from the sensors (input terminals A-1 through E-1 shall be inhibited from presenting an alarm condition between output terminal S-1-I and its return, between S-1-D and its return and between L-2-1 and L-2-2. Tamper alarms (input terminals A-3 through F-3 and S-8 shall not be inhibited and shall be presented to the output terminals as described in 3.6.1.5.3. Duress alarms (input terminal F-1) shall not be inhibited between S-1-I and its return and between S-1-D and its return and shall be presented to these terminals as described in 3.6.1.5.3, but shall be inhibited from appearing between L-2-1 and L-2-2. A standard output alarm condition in the low impedance state shall be presented between output terminal S-2 and its return. When initiated, this low impedance state shall be maintained for a minimum period greater than 4.5 seconds but less than 6.0 seconds, even if the operating mode switch is turned to the SECURE position. In this mode only, unregulated 24 Vdc, at a maximum current of 50 mA, shall be supplied to each of the terminals A-7 through E-7.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:10Z
Check the source to verify that this is the currentversion before use.

3.6.1.6.2 TEST/RESET. Alarm outputs between terminal S-1-I and its return and between S-1-D and its return shall not be inhibited but shall be instantaneously presented across the terminal pairs as described in 3.6.1.5.3.1 and 3.6.1.5.3.2. The alarm output presented between S-1-D and its return, however, shall not latch as described in 3.6.1.5.3.2. This alarm output shall be maintained for the same period of time (10 ±2 seconds) as the alarm presented between S-1-I and its return. All alarms presented between L-2-1 and L-2-2 shall be inhibited. A standard output alarm condition in the low impedance state, shall be presented between output terminal S-2 and its return. When initiated, this low impedance state shall be maintained for a period greater than 4.5 seconds but less than 6.0 seconds, even if the SECURE/ACCESS switch is turned to the SECURE position. The audible buzzer shall sound in this mode only when an alarm condition is presented at output terminal S-1-I. The entrance alarm time delay circuit shall be reset immediately upon rotating the operating mode switch to the TEST/RESET position from the SECURE position.

3.6.1.6.3 SECURE. A standard output alarm condition in the high impedance state, shall appear between output terminal S-2 and its return, providing the low impedance state initiated as discussed above has been maintained for the specified period. Intrusion alarms shall be delayed from appearing between terminal S-1-D and its return and between L-2-1 and L-2-2 for the adjustable entrance time delay (screwdriver adjustable). Intrusion alarms shall instantaneously appear between terminal S-1-I and its return for a 10 ±2 second time period. If the operating mode switch is moved to the TEST/RESET position prior to expiration of the entrance time delay, the latched alarm that would normally appear between S-1-D and its return and between L-2-1 and L-2-2 shall be inhibited.

3.6.1.7 Transmitter resync switch. When transmitter resync switch S2 is pushed, Terminal S-7 shall be placed at the same potential as terminal S-4.

3.6.1.8 Alarm line supervision. The circuit card connector pins provided in the control unit shall provide individual electrical access to all intrusion, duress, and tamper alarm input lines and their associated signal return lines. Power (20 ±1 Vdc), shall also be available at these pins, along with two pins which are connected one to each of two terminals, in close proximity to the circuit card connector. These terminals shall be in series with a control unit tamper alarm input so that when the

Source: https://assist.dla.mil -- Downloaded: 2026¹0-26T04:10Z
Check the source to verify that this is the current version before use.

terminals are not jumpered and the connector pins are not shorted, a tamper alarm is presented to the status processor. With these terminals jumpered and without a circuit card installed, the control unit shall perform as specified herein.

3.6.1.9 Tamper/alarm printed wiring board assembly. The tamper/alarm printed wiring board assembly, A1 through A6, shall be in accordance with Top Assembly TA13220E4213 and as specified herein. The tamper/alarm wiring board assembly shall have the following electrical characteristics:

(a) With a supply voltage of 19.9 ±0.1 Vdc, an impedance change from 20k ohms ±1 percent to greater than 100k ohms for a duration of 65 ±2 ms at either the tamper or intrusion inputs shall not result in an alarm output duration greater than 10 ±0.05 ms.

(b) With a supply voltage of 19.9 ±0.1 Vdc, an impedance change from 20k ohms ±1 percent to greater than 100k ohms for a duration of 135 ±15 ms at either the tamper or intrusion inputs shall result in an alarm output duration greater than 20 ±0.1 ms.

3.6.1.10 Voltage monitor printed wiring board assembly. The voltage monitor printed wiring board assembly, A7, shall be in accordance with Top Assembly TA13220E4214 and as specified herein. The voltage monitor printed wiring board assembly shall have the following electrical characteristics:

(a) With a supply voltage of 19.9 ±0.1 Vdc, an impedance change from 20k ohms ±1 percent to greater than 100k ohms for 65 ±2 ms at either the door or telephone dialer tamper inputs shall not result in an alarm output duration greater than 10 ±0.05 ms.
(b) With a supply voltage of 19.9 ±0.1 Vdc, an impedance change from 20k ohms ±1 percent to greater than 100k ohms for a duration of 135 ±15 ms at either the door or telephone dialer tamper inputs shall result in an alarm output duration greater than 20 ±0.1 ms at greater than 18 Vdc.
(c) With a supply voltage of 19.9 ±0.1 Vdc and an input voltage at terminal pin 14 of 4 to 5.5 Vdc, the alarm output voltage shall be less than 1 Vdc.
(d) With a supply voltage of 19.9 ±0.1 Vdc and an input voltage at terminal pin 14 of less than 1 Vdc, the alarm output voltage shall be greater than 18 Vdc.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:10Z
Check the source to verify that this is the currentversion before use.

(e) With a supply voltage equal to or greater than 19 and equal to or less than 21 Vdc, the alarm output voltage shall be less than 1 Vdc.

(f) With a supply voltage equal to or less than 18.6 or equal to or greater than 21.4 Vdc, the alarm output voltage shall be greater than 12 Vdc.

3.6.1.11 Audible alarm printed wiring board assembly. The audible alarm printed wiring board assembly, A8, shall be in accordance with Top Assembly TA13220E4215 and as specified herein. The audible alarm printed wiring board assembly shall have the following electrical characteristics with 19.9 ±0.1 Vdc applied to terminal pin 9, and 4.9 ±0.1 Vdc at terminal pin 10. Unless otherwise stated, all voltages and impedances are referenced to ground (terminal pin 1).

3.6.1.11.1 Access timer requirements. After an input pulse between 18 and 20 Vdc of 20 ±0.5 ms and greater duration on terminal pins 15 or 16, an impedance of between 0 to 1 ohm shall exist between terminal pins 2 and 3 for a duration of 5.5 ±1.5 seconds. An impedance greater than 100k ohms shall exist between terminal pins 2 and 3 at any time greater than 7 seconds subsequent to input pulse termination.

3.6.1.11.2 Nonlatch timer requirements. After an input pulse between 18 and 20 Vdc of 20 ±0.5 ms and greater duration on terminal pin 5, an impedance of between 0 and 100 ohms shall exist between terminal pins 1 and 6 for a duration of 10 ±2 seconds. An impedance greater than 100k ohms shall exist between terminal pins 1 and 6 at any time greater than 12 seconds subsequent to imput pulse termination.

3.6.1.11.3 Audible alarm inhibit requirements. With 18 to 20 Vdc applied to terminal pin 11 and an impedance greater than 100k ohms between terminal pins 1 and 12, an output voltage of between 18 and 20 Vdc shall exist at terminal pin 8. With 18 to 20 Vdc supplied to terminal pin 11 and an impedance of between 0 and 100 ohms between terminal pins 2 and 12, an output voltage between 0 and 1 Vdc shall exist at terminal pin 8.

3.6.1.11.4 Tamper alarm requirements. With 18 to 20 Vdc applied to terminal pin 11 and an impedance greater than 100k ohms between terminal pins 1 and 12, an output voltage between 18 and 20 Vdc shall exist at terminal pin 8. With 0 to 1 Vdc applied to terminal pin 11, an output voltage between 0 and 1 Vdc shall exist at terminal pin 8.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

3.6.1.11.5 Sonalert driver requirements. With a voltage between 4 to 5 Vdc applied to terminal pin 14, an impedance between 0 and 100 ohms shall exist between terminal pins 1 and 13. With a voltage less than 1 Vdc applied to terminal pin 14, an impedance greater than 100k ohms shall exist between terminal pins 1 and 13.

3.6.1.11.6 Access command requirements. With a voltage between 18 to 20 Vdc applied to terminal pin 16, an impedance between 0 and 1 ohm shall exist between terminal pins 17 and 18. With less than 1 Vdc applied to terminal pin 16, an impedance greater than 100k ohms shall exist between terminal pins 17 and 18.

3.6.1.12 Instantaneous alarm printed wiring board assembly. The instantaneous alarm printed wiring board assembly, A9, shall be in accordance with Top Assembly TA13220E4216 and as specified herein. The instantaneous alarm printed wiring board assembly shall have the following electrical characteristics with 4.9 ±0.1 Vdc applied to terminal pin 11, and 19.9 ±0.1 Vdc applied to terminal pins 3 and 17. Unless otherwise stated, all voltages and impedances are referenced to ground (terminal pin 1).

3.6.1.12.1 Instantaneous timer requirements. An alarm input pulse at either terminal pins 8, 9, or 10 of between 18 to 20 Vdc for a duration of 75 ±1 ms and greater shall cause the following impedance conditions to exist:

Pin	to	Pin	Impedance
1		R	0 to 100 ohms
16		18	0 to 1 ohm
4		B	Greater than 100k ohms
14		U	Greater than 100k ohms

These conditions shall remain until the alarm input terminates and for a minimum additional duration of 10 ±2 seconds. Subsequent to the termination of the alarm input pulse and subsequent to a minimum additional duration of 12 seconds, the following impedance conditions shall exist:

Pin	to	Pin	Impedance
1		R	Greater than 100k ohms
4		N	0 to 1 ohm
14		U	0 to 1 ohm
15		13	0 to 1 ohm

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:10Z
Check the source to verify that this is the current version before use.

3.6.1.12.2 Nonlatch alarm output requirements. With 18 to 20 Vdc applied to terminal pin D (reference terminal pin C), an impedance greater than 100k ohms shall exist between terminal pins 4 and B. With the voltage removed, an impedance between 0 and 1 ohm shall exist between terminal pins 4 and N.

3.6.1.12.3 Exit delay timer requirements. An alarm input at terminal pin 13 between 4 and 5 Vdc with a duration of 20 ±0.5 ms shall result in a delayed output pulse at terminal pin S between 4 and 5 Vdc with a minimum duration of 25 ±1 ms. The output pulse shall occur at 10 ±2 seconds for the minimum exit delay setting and at 80 to 150 seconds for the maximum exit delay setting. For alarm inputs exceeding the exit delay time, the duration of the delayed output at terminal pin S shall be equal to the difference (±5 percent) between the input duration and the exit delay time.

3.6.1.12.4 Exit delay alarm inhibit. After the expiration of the exit delay timer, a voltage between 0 and 1 Vdc shall be present at terminal pin 12. With 15 to 20 Vdc applied to terminal pin 9, the voltage at terminal pin 12 shall be between 4 and 5 Vdc.

3.6.1.13 Latched alarm printed wiring board assembly. The latched alarm printed wiring board assembly, A10, shall be in accordance with Top Assembly TA13220E4217 and as specified herein. The latched alarm printed wiring board assembly shall have the following electrical characteristics with 4.9 ±0.1 Vdc applied to terminal pin 5. Unless otherwise stated, all voltages and impedances are referenced to ground (terminal pin 1).

3.6.1.13.1 Entrance timer requirements. An alarm input at terminal pin 3 between 4 and 5 Vdc with a duration of 20 ±0.5 ms shall result in a delayed output pulse at terminal pin 18 between 4 and 5 Vdc with a minimum duration of 25 ±1 ms. The output pulse shall occur at 10 ±2 seconds for the minimum entrance delay setting and at 80 to 150 seconds for the maximum entrance delay setting. For alarm inputs exceeding the entrance delay time, the duration of the delayed output at terminal pin 18 shall be equal to the time difference (±5 percent) between the input duration and the entrance delay time.

3.6.1.13.2 Audible alarm output requirements. With 18 to 20 Vdc applied to either of terminal pins P or H, and the entrance delay timer reset, the voltage at terminal pin 4 shall be between 4 and 5 Vdc. Once an alarm input is received at terminal pin 3 and a delayed output has occurred at terminal pin 18, the voltage at terminal pin 4 and test point TP1 shall be latched at 0 to 1 Vdc. Momentary removal of the voltage at terminal pins P and H shall cause the latch to reset, the voltage at TP1 to be greater than 18 Vdc, and the voltage at terminal pin 4 to be between 4 and 5 Vdc.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

3.6.1.13.3 Duress alarm requirements. With 18 to 20 Vdc applied to terminal pin L, and the entrance delay timer reset, the impedance between terminal pins 8 and 10 and between terminal pins 8 and 9 shall be between 0 and 1 ohm. An alarm input at terminal pin 6 between 4 and 5 Vdc with a duration of 20 ±0.5 ms, shall immediately cause the duress output relay to latch ON and an impedance greater than 100k ohms to be present between terminal pins 8 and 10. Momentary removal of the voltage at terminal pin L shall cause the latch to reset and the impedance between terminal pins 8 and 10 and between terminal pins 8 and 9 to be between 0 and 1 ohm.

3.6.1.13.4 Tamper and intrusion alarm requirements. With 18 to 20 Vdc applied to terminal pin L, and the entrance delay timer and duress latch reset, the impedance between terminal pins 8 and 10 shall be between 0 and 1 ohm. Once an alarm input is received at terminal pin 3 and a delayed output has occurred at terminal pin 18, the impedance between terminal pins 8 and 10 shall be latched at greater than 100k ohms. Momentary removal of the voltage at terminal pin L shall cause the latch to reset and the impedance between terminal pins 8 and 10 to be between 0 and 1 ohm.

3.6.1.13.5 Master relay requirements. With 18 to 20 Vdc applied to terminal pin 17, the impedance between terminal pins 12 and 13 and between terminal pins 10 and 15 shall be greater than 100k ohms, and the impedance between terminal pins 1 and R shall be between 0 and 1 ohm. With the voltage removed from terminal pin 17, the impedance between terminal pins 11 and R, terminal pins 12 and 14, and terminal pins 15 and 16, shall be between 0 and 1 ohm.

3.6.1.14 Alarm ID printed wiring board assembly. The alarm ID printed wiring board assembly, A11, shall be in accordance with Top Assembly TA13220E4218 and as specified herein. The alarm ID printed wiring board assembly shall have the following electrical characteristics with supply voltages of 19.9 ±0.1 Vdc and 4.9 ±0.1 Vdc.

(a) With no-alarm inputs, momentary application of 20 Vdc to terminal pin 10 or momentary grounding of terminal pin 8, shall cause a reset of all latches and shall cause all LED driver transistors to turn OFF.

(b) Not less than 5 seconds after a reset pulse on terminal pin 8 or terminal pin 10, an alarm input greater than 18 Vdc with a duration of 15 ±1 ms presented to any input terminal pin 2, 4, 12, 16, or 18, shall cause the corresponding latches to latch and the corresponding LED driver transistors to turn ON.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

(c) Any alarm input greater than 18 Vdc which occurs at any input terminal pin 2, 4, 12, 14, 16, or 18 within 0.8 ±0.1 seconds subsequent to the receipt of the leading edge of an alarm input on any input terminal pin shall cause the corresponding latches to latch and the corresponding LED driver transistors to turn ON.

(d) Any alarm input greater than 18 Vdc which occurs at input terminal pin 2, 4, 12, 14, 16, or 18 after 1.5 ±0.2 seconds subsequent to receipt of the leading edge of an alarm input on any input terminal pin shall not affect the status of the latches or LED driver transistors. Latches which are set shall remain set; latches which are reset shall remain reset; momentary activation of non-activated LED driver transistors shall be cause for rejection.

3.6.1.15 LED printed wiring board assembly. The LED printed wiring board assembly, A12, shall be in accordance with Top Assembly TA13220E4219 and as specified herein. The LED printed wiring board assembly shall have all light emitting diodes mounted so that the bases of the LED's are parallel to the plane of the card and with a supply voltage of 19.9 ±0.1 Vdc and 2.2k ohm ±10 percent series load resistors, all LED's of the card shall be lighted with the same apparent uniform brightness and intensity.

3.7 Electromagnetic susceptibility. The electromagnetic susceptibility characteristics of the control unit shall conform to MIL-STD-461, Methods RS03, CS01, CS02, and CS06, as modified in 3.7.1 and 3.7.2.

3.7.1 Radiated. Radiated susceptibility shall be conducted in an electric field of 5 volts per meter (V/m) for both steady-state and keyed conditions of the EMI field generating equipment; this field applies to frequencies up to 500 megahertz (MHz). Above 500 MHz, the electric field is 1 V/m.

3.7.2 Conducted. Conducted interference signals shall be applied to the power lines, to the lines attached to the duress tamper and duress alarm inputs, and to the lines to one other tamper input and one other alarm input selected at random as follows:

(a) Continuous wave, 30 Hz to 400 kilohertz (kHz).
(b) Interfering spike voltage, 10 spikes 5 to 10 microseconds wide at a repetition rate of 7 to 10 pulses per second, on input power leads only. Random phasing of the spikes with respect to the power line frequency shall be employed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

3.8 Reliability. The control unit shall demonstrate a Mean-Time-Between-Failure (MTBF) of 10,000 hours when tested as specified in 4.6.2.7.

3.9 Identification marking. The control unit and major components shall be identified in accordance with MIL-STD-130.

3.10 Government-loaned property. Unless otherwise specified (see 6.2), the following property in the quantity indicated will be loaned by the Government (see 6.8), for testing major components furnished separately.

Item No.	Description	Identification	Quantity
1.	Control Unit, Alarm Set	C-9412/FSS-9(V) NSN 6350-00-228-2735	1

3.11 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts, except for adjustment, and without selection for fit or performance.

3.12 Workmanship. Workmanship shall be in accordance with MIL-STD-454, Requirement 9, except soldering. These requirements are applicable to wiring, welding, brazing, plating, riveting, finishes, machine operations, screw assemblies, and freedom of parts from burrs, sharp edges, or any other damage or defect that could make the part (or equipment) unsuitable for the purpose intended.

3.12.1 Soldering. All soldering shall be performed in accordance with MIL-STD-454, Requirement 5.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.1.2 Disassembly inspection. Failure of any examination or test by the preproduction model shall be cause for disassembly, in the presence of a Government representative, of the preproduction model to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the preproduction model. Reassembly with replacement parts and retesting shall be the responsibility of the contractor. Government-loaned property shall be excluded from the requirements of this paragraph.

4.1.3 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the quality assurance provisions (QAP) shown on the drawings. The drawings specify the characteristics requiring QAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.4.4 and 6.5).

4.2 Classification of inspections. The control unit shall be inspected as specified herein. Individual testing of major components installed in the control unit is not required. Major components furnished separately shall be individually inspected as specified herein. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of packaging (see 4.8).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction control units and major components furnished separately, shall be examined as specified in 4.6.1. Presence of one or more defect shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.3.2 Tests.

4.3.2.1 Complete control units. Complete control units shall be subjected to the tests marked "X" in Column 1 of Table I. Individual testing of major components of the complete control unit is not required. Failure of any test shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.3.2.2 Major components furnished separately. Major components furnished separately shall be subjected to tests marked "X" in Column 1 of Tables II through IX. Failure of any test shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.4 Initial production inspection. When specified (see 3.3), one or more initial production control units will be selected at random by the Government from the control units being produced by production tooling, and will be examined as specified in 4.6.1 and subjected to the tests marked "X" in Column 2 of Table I to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production control unit shall not exclude the remaining control units from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any or all other tests contained in this specification as part of the initial production test, and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of an initial production control unit to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production control units and shall be cause for refusal by the Government to continue acceptance of production control units until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

Government on control units previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all control units accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105, Inspection Level II. Sampling for tests shall be in accordance with MIL-STD-105, Inspection Level S-2.

4.5.2 Examination.

4.5.2.1 Individual. Each control unit and major component furnished separately, shall be examined for the critical defect specified in 4.6.1. Presence of the defect shall be cause for rejection.

4.5.2.2 Samples. Samples selected in accordance with 4.5.1 shall be examined for the major defects specified in 4.6.1. AQL shall be 0.4 percent defective.

4.5.3 Tests.

4.5.3.1 Individual.

4.5.3.1.1 Control units. Each control unit shall be subjected to the tests marked "X" in Column 3 of Table I. Individual testing of major components of the complete control unit is not required. Failure of any test shall be cause for rejection.

4.5.3.1.2 Major components furnished separately. Each major component furnished separately shall be subjected to the tests marked "X" in Column 2 of Tables II through IX. Failure of any test shall be cause for rejection.

4.5.3.2 Samples.

4.5.3.2.1 Control units. Samples selected in accordance with 4.5.1 shall be subjected to the tests marked "X" in Column 4 of Table I, except that not more than three sample units shall be subjected to the humidity test specified in 4.6.2.5.6. Individual testing of major components of complete control unit is not required. AQL shall be 2.5 percent defective.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

4.5.3.2.2 Major components furnished separately. Samples selected in accordance with 4.5.1 shall be subjected to the tests marked "X" in Column 3 of Tables II through IX, except that not more than three samples of the major component shall be subjected to the humidity test specified in 4.6.2.5.6. AQL shall be 2.5 percent defective.

4.6 Inspection procedure.

4.6.1 Examination. The control unit and major components furnished separately shall be examined as specified herein for the following defects:

Critical

1. Welding burn through.

Major

101. Any part (or component) not in accordance with the QAP requirements as shown on the drawings.
102. Any dimension, other than those identified on the QAP requirements shown on the drawings, not as specified.
103. Components missing or not as specified.
104. Assembly not correct.
105. Electromagnetic interference suppression components and network not identical to the approved model.
106. Identification marking or special marking missing, incomplete, or illegible.
107. Workmanship not as specified.

4.6.2 Tests.

4.6.2.1 Test conditions. Unless otherwise specified herein, tests shall be conducted at prevailing ambient temperature and humidity conditions. Prior to testing, each control unit and major components furnished separately, except batteries, shall be subjected to a

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

72-hour burn-in. Burn-in control units in accordance with Appendix Test Method 5. The power supply of each control unit shall be burned-in in accordance with Appendix Test Method 6. The control unit and its power supply may be burned-in separately with Test Method 5 being applicable to the control unit less supply and Test Method 6 being applicable to the supply. Major components when furnished separately shall be activated nine times each 24 hours to demonstrate all functions, and shall be tested in accordance with the applicable test methods in the Appendix, and comply with their requirements. All failures shall be recorded and analyzed for cause of failure. Corrective action shall be taken when appropriate. Copies of these failure reports shall be forwarded to the contracting officer. Units for testing shall be selected from those units having successfully completed the 72-hour burn-in.

4.6.2.2 Test schedule.

TABLE I. Test schedule for complete control unit.

Preproduction	Initial production	Quality conformance		Test	Requirement paragraph	Test paragraph
		Individual	Sample			
1	2	3	4	5	6	7
X	X	X	-	Functional	3.6.1.1 and 3.6.1.3 thru 3.6.1.8	4.6.2.4
X	X	-	-	Standby power	3.6.1.4	4.6.2.4.1
X	X	X	-	Tamper switch tolerance	3.6.1.2.1	4.6.2.4.2
X	X	-	-	Mode switch life	3.6.1.6	4.6.2.4.3
X	X	-	X	Low temperature	3.5.1(a), 3.5.2(a)	4.6.2.5.2
X	X	-	X	High temperature	3.5.1(a), 3.5.2(a)	4.6.2.5.3
X	X	-	-	Shock	3.5.2(c)	4.6.2.5.4
X	X	-	-	Vibration	3.5.2(d)	4.6.2.5.5
X	X	-	X	Humidity	3.5.1(b), 3.5.2(b)	4.6.2.5.6
X	X	-	-	Electromagnetic susceptibility	3.7	4.6.2.6
X	X	-	-	Reliability	3.8	4.6.2.7

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

TABLE II. Test schedule for power supply assembly.

Preproduction	Quality Conformance		Test	Requirement paragraph	Test paragraph
	Individual	Sample			
1	2	3	4	5	6
X	X	-	Functional	3.6.1.3	4.6.2.4.4
X	-	X	Low temperature 1/	3.5.1(a), 3.5.2(a)	4.6.2.5.2
X	-	X	High temperature 1/	3.5.1(a), 3.5.2(a)	4.6.2.5.3
X	-	-	Shock 1/	3.5.2(c)	4.6.2.5.4
X	-	-	Vibration	3.5.2(d)	4.6.2.5.5
X	-	X	Humidity 1/	3.5.1(b), 3.5.2(b)	4.6.2.5.6
X	-	-	Electromagnetic susceptibility 1/	3.7	4.6.2.6

1/ Place power supply assembly in Government-loaned control unit or equivalent to simulate normal environment when installed in control unit.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:10Z
Check the source to verify that this is the currentversion before use.

TABLE III. Test schedule for tamper/alarm printed wiring board assembly.

Preproduction	Quality Conformance		Test	Requirement paragraph	Test paragraph
	Individual	Sample			
1	2	3	4	5	6
X	X	-	Functional	3.6.1.9	4.6.2.4.5
X	-	X	Low temperature	3.5.1(a), 3.5.2(a)	4.6.2.5.2
X	-	X	High temperature	3.5.1(a), 3.5.2(a)	4.6.2.5.3
X	-	-	Shock 1/	3.5.2(c)	4.6.2.5.4
X	-	-	Vibration 1/	3.5.2(d)	4.6.2.5.5
X	-	X	Humidity 1/	3.5.1(b), 3.5.2(b)	4.6.2.5.6

1/ Place tamper/alarm printed wiring board assembly in Government-loaned control unit or equivalent to simulate normal environment when installed in control unit. Board may be removed from control unit to perform required electrical tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

TABLE IV. Test schedule for voltage monitor printed wiring board assembly.

Preproduction	Quality Conformance		Test	Requirement paragraph	Test paragraph
	Individual	Sample			
1	2	3	4	5	6
X	X	-	Functional	3.6.1.10	4.6.2.4.6
X	-	X	Low temperature	3.5.1(a), 3.5.2(a)	4.6.2.5.2
X	-	X	High temperature	3.5.1(a), 3.5.2(a)	4.6.2.5.3
X	-	-	Shock 1/	3.5.2(c)	4.6.2.5.4
X	-	-	Vibration 1/	3.5.2(d)	4.6.2.5.5
X	-	X	Humidity 1/	3.5.1(b) 3.5.2(b)	4.6.2.5.6

1/ Place voltage monitor printed wiring board assembly in Government-loaned control unit or equivalent to simulate normal environment when installed in control unit. Board may be removed from control unit to perform required electrical tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

TABLE V. Test schedule for audible alarm printed wiring board assembly.

Preproduction	Quality Conformance		Test	Requirement paragraph	Test paragraph
	Individual	Sample			
1	2	3	4	5	6
X	X	-	Functional	3.6.1.11	4.6.2.4.7
X	-	X	Low temperature	3.5.1(a), 3.5.2(a)	4.6.2.5.2
X	-	X	High temperature	3.5.1(a), 3.5.2(a)	4.6.2.5.3
X	-	-	Shock 1/	3.5.2(c)	4.6.2.5.4
X	-	-	Vibration 1/	3.5.2(d)	4.6.2.5.5
X	-	X	Humidity 1/	3.5.1(b), 3.5.2(b)	4.6.2.5.6

1/ Place the audible alarm printed wiring board assembly in Government-loaned control unit or equivalent to simulate normal environment when installed in control unit. Board may be removed from control unit to perform required electrical tests.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:10Z
Check the source to verify that this is the current version before use.

TABLE VI. Test schedule for instantaneous alarm printed wiring board assembly.

Preproduction	Quality Conformance		Test	Requirement paragraph	Test paragraph
	Individual	Sample			
1	2	3	4	5	6
X	X	-	Functional	3.6.1.12	4.6.2.4.8
X	-	X	Low temperature	3.5.1(a), 3.5.2(a)	4.6.2.5.2
X	-	X	High temperature	3.5.1(a), 3.5.2(a)	4.6.2.5.3
X	-	-	Shock 1/	3.5.2(c)	4.6.2.5.4
X	-	-	Vibration 1/	3.5.2(d)	4.6.2.5.5
X	-	X	Humidity 1/	3.5.1(b), 3.5.2(b)	4.6.2.5.6

1/ Place the instantaneous alarm printed wiring board assembly in Government-loaned control unit or equivalent to simulate normal environment when installed in control unit. Board may be removed to perform required electrical tests.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:10Z
Check the source to verify that this is the current version before use.

TABLE VII. Test schedule for latched alarm printed wiring board assembly.

Preproduction	Quality Conformance		Test	Requirement paragraph	Test paragraph
	Individual	Sample			
1	2	3	4	5	6
X	X	-	Functional	3.6.1.13	4.6.2.4.9
X	-	X	Low temperature	3.5.1(a), 3.5.2(a)	4.6.2.5.2
X	-	X	High temperature	3.5.1(a), 3.5.2(a)	4.6.2.5.3
X	-	-	Shock 1/	3.5.2(c)	4.6.2.5.4
X	-	-	Vibration 1/	3.5.2(d)	4.6.2.5.5
X	-	X	Humidity 1/	3.5.1(b), 3.5.2(b)	4.6.2.5.6

1/ Place latched alarm printed wiring board assembly in Government-loaned control unit or equivalent to simulate normal environment when installed in control unit. Board may be removed from control unit to perform required electrical tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

TABLE VIII. Test schedule for alarm I.D. printed wiring board assembly.

Preproduction	Quality Conformance		Test	Requirement paragraph	Test paragraph
	Individual	Sample			
1	2	3	4	5	6
X	X	-	Functional	3.6.1.14	4.6.2.4.10
X	-	X	Low temperature	3.5.1(a), 3.5.2(a)	4.6.2.5.2
X	-	X	High temperature	3.5.1(a) 3.5.2(a)	4.6.2.5.3
X	-	-	Shock 1/	3.5.2(c)	4.6.2.5.4
X	-	-	Vibration 1/	3.5.2(d)	4.6.2.5.5
X	-	X	Humidity 1/	3.5.1(b), 3.5.2(b)	4.6.2.5.6

1/ Place alarm I.D. printed wiring board assembly in Government-loaned control unit or equivalent to simulate normal environment when installed in control unit. Board may be removed from control unit to perform required electrical tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

TABLE IX. Test schedule for LED printed wiring board assembly.

Preproduction	Quality Conformance		Test	Requirement paragraph	Test paragraph
	Individual	Sample			
1	2	3	4	5	6
X	X	-	Functional	3.6.1.15	4.6.2.4.11
X	-	X	Low temperature	3.5.1(a), 3.5.2(a)	4.6.2.5.2
X	-	X	High temperature	3.5.1(a), 3.5.2(a)	4.6.2.5.3
X	-	-	Shock 1/	3.5.2(c)	4.6.2.5.4
X	-	-	Vibration 1/	3.5.2(d)	4.6.2.5.5
X	-	X	Humidity 1/	3.5.1(b), 3.5.2(b)	4.6.2.5.6

1/ Place LED printed wiring board assembly in Government-loaned control unit or equivalent to simulate normal environment when installed in control unit. Board may be removed from control unit to perform required electrical tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

4.6.2.3 Test methods. Tests shall be conducted in accordance with the Appendix Test Methods and as specified herein. Calibration of the test equipment shall be in accordance with MIL-C-45662.

4.6.2.4 Functional. The control unit shall be tested in accordance with Appendix Test Method 1. Nonconformance to 3.6.1.1 and 3.6.1.3 through 3.6.1.8 shall constitute failure of this test.

4.6.2.4.1 Standby power. The control unit shall be tested in accordance with Appendix Test Method 2. Nonconformance to 3.6.1.4 shall constitute failure of this test.

4.6.2.4.2 Tamper switch tolerance. The control unit shall be tested in accordance with Appendix Test Method 3. Nonconformance to 3.6.1.2.1 shall constitute failure of this test.

4.6.2.4.3 Mode switch life. The control unit shall be tested in accordance with Appendix Test Method 4. Nonconformance to 3.6.1.6 shall constitute failure of this test.

4.6.2.4.4 Power supply assembly. The power supply assembly shall be tested in accordance with Appendix Test Method 6. Nonconformance to 3.6.1.3 shall constitute failure of this test.

4.6.2.4.5 Tamper/alarm printed wiring board assembly. The tamper/alarm printed wiring board assembly shall be tested in accordance with Appendix Test Method 7. Nonconformance to 3.6.1.9 shall constitute failure of this test.

4.6.2.4.6 Voltage monitor printed wiring board assembly. The voltage monitor printed wiring board assembly shall be tested in accordance with Appendix Test Method 8. Nonconformance to 3.6.1.10 shall constitute failure of this test.

4.6.2.4.7 Audible alarm printed wiring board assembly. The audible alarm printed wiring board assembly shall be tested in accordance with Appendix Test Method 9. Nonconformance to 3.6.1.11 shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

4.6.2.4.8 Instantaneous alarm printed wiring board assembly. The instantaneous alarm printed wiring board assembly shall be tested in accordance with Appendix Test Method 10. Nonconformance to 3.6.1.12 shall constitute failure of this test.

4.6.2.4.9 Latched alarm printed wiring board assembly. The latched alarm printed wiring board assembly shall be tested in accordance with Appendix Test Method 11. Nonconformance to 3.6.1.13 shall constitute failure of this test.

4.6.2.4.10 Alarm I.D. printed wiring board assembly. The alarm I.D. printed board assembly shall be tested in accordance with Appendix Test Method 12. Nonconformance to 3.6.1.14 shall constitute failure of this test.

4.6.2.4.11 LED printed wiring board assembly. The LED printed wiring board assembly shall be tested in accordance with Appendix Test Method 12. Nonconformance to 3.6.1.15 shall constitute failure of this test.

4.6.2.5 Environmental. The control unit and major components furnished separately, shall be subjected to the following sequence of environmental tests of MIL-STD-810.

4.6.2.5.1 Functional. Functional testing required for environmental testing shall be conducted as specified in 4.6.2.4, as applicable.

4.6.2.5.2 Low temperature. Low temperature Test Method 502.1, Procedure I, storage temperature -30° F for not less than 48 hours, equipment operating temperature -20° F.

4.6.2.5.3 High temperature. High temperature Test Method 501.1, Procedure II, except that the non-operating temperature shall be +165° F; equipment operating temperature shall be +150° F.

4.6.2.5.4 Shock.

(a) Basic design: Test Method 516.2, Procedure I. Transportation conditions (without battery). Sawtooth shock pulse, peak value 20 g's, duration 11 ms.
(b) Bench handling. Test Method 516.2, Procedure V. Equipment not operating (with battery).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

4.6.2.5.5 Vibration. Test Method 514.2, Procedure X, Curve AW, transportation conditions (without battery).

4.6.2.5.6 Humidity test. Test Method 507.1, Procedure I with the following exceptions:

(a) Step 5 shall be for 5 cycles (not less than 120 hours). Take measurements at the completion of each cycle.
(b) During Steps 2 and 3, the relative humidity shall be maintained at 95 percent or greater.
(c) During Step 4, the relative humidity shall be maintained at 92 percent or greater.
(d) Measurement shall commence 1 hour and 15 minutes prior to the completion of each 24 hour cycle. The doors of the control unit shall not be opened during measurements. The control unit may be removed from the environmental chamber for measurement; however, unsuccessful completion of measurements or unsuccessful return of the control unit to the test chamber within 1 hour and 15 minutes subsequent to the initiation of measurements shall constitute failure of the test.
(e) The control unit may be opened to remove printed wiring boards furnished separately, for testing. Testing shall commence 1 hour plus or minus 15 minutes prior to the completion of each 24 hour cycle. Nonconformance to the appropriate functional requirements for the printed wiring boards when tested in accordance with the applicable test methods in the Appendix and failure to return the printed wiring board to the test chamber within 1 hour plus or minus 15 minutes subsequent to initiation of testing shall constitute failure of the test.

4.6.2.5.7 Failure criteria. Nonconformance to 3.5 shall constitute failure of these tests.

4.6.2.6 Electromagnetic susceptibility. The control unit and power control assembly furnished separately, shall be subjected to the tests of MIL-STD-461 as follows:

4.6.2.6.1 Radiated susceptibility. Test Method RS03. Electric field requirements of RS03 shall be modified and limited as specified in 3.7.1. All leads (sensor, alarm, and power) external to the enclosure shall be exposed to the electric field. The leads shall be not less than 4 feet in length.

4.6.2.6.2 Conducted susceptibility. Test Methods CS01, CS02, and CS06, as modified in 3.7.2.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:10Z
Check the source to verify that this is the current version before use.

4.6.2.6.3 Failure criteria. The contractor shall furnish the contracting officer a report of the tests required by MIL-STD-461. Disapproval of the test report shall constitute failure of this test.

4.6.2.7 Reliability. Using an upper test MTBF (θ_0) of 10,000 hours and a lower test MTBF (θ_1) of 5000 hours the control unit shall be tested as specified herein, with "accept" and "reject" criteria in accordance with MIL-STD-781, Test Plan VIIIC. Continue testing until either an "accept" or "reject" decision is reached. A failure is defined as any malfunction which causes or may cause one or more of the following.

(a) Failure to commence operation, cessation of operation, or degradation of performance below the designated levels.
(b) Serious damage to the system.
(c) Serious personnel safety hazards.

4.6.2.7.1 Test procedure. The tests shall be conducted under ambient laboratory conditions, without temperature, humidity, or vibration cycling. Equipment shall be energized at the primary power input specified herein. Operation shall be continuous throughout the test with all circuits exercised nine times each 24-hour period. The parameters to be measured shall be in accordance with Test Method 1. The test shall commence after installation and a burn-in period of 72 hours, as specified in 4.6.2.1. Parts or components of known limited life, such as batteries, shall have a life stipulated prior to test. Failures of these parts occurring after the end of the stipulated period are nonrelevant, but any dependent failures caused thereby are counted as relevent. Failures chargeable to workmanship deficiencies shall be counted as relevant. Alarms not deliberately excited as part of the equipment test routine shall be reported and cause of the alarm identified, if practicable. This accounting shall be maintained separately from the failure accounting. Redundant elements, such as indicator lamps shall not be repaired nor counted as a failure until the control unit has failed to perform its intended function. Preventive maintenance procedure specified for the equipment during normal operation shall be performed during the tests. No additional preventive maintenance shall be permitted during the tests.

4.6.2.7.2 Failure criteria. Nonconformance to 3.8 shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

4.7 Inspection comparison. The Government may select control units at any time during the contract production period and subject these control units to the examination and tests specified in 4.4 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on control units selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production control units. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison control unit to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison control unit and shall be cause for refusal by the Government to continue acceptance of production control units until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on control units previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all control units accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.8 Inspection of packaging.

4.8.1 Quality conformance inspection of pack.

4.8.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:10Z
Check the source to verify that this is the current version before use.

4.8.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.8.1.3 Examination. Samples selected in accordance with 4.8.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

108.	Materials, methods and containers not as specified for Level A or Level B. Each incorrect material, method, or container shall be considered one defect.
109.	Batteries not preserved and packed separate from the control unit as specified for Level A, Level B, or Level C.
110.	Keys not placed with the control unit as specified for Level A, Level B, or Level C.
111.	Major components furnished separately not preserved and packed as specified for Level A or Level B.
112.	Components of unlike description consolidated or packed together for Level A or Level B.
113.	Quantities packed together exceed the size or weight limitations of the box for Level A or Level B.
114.	Strapping not as specified for Level A.
115.	Marking missing, illegible, incorrect, or incomplete for Level A, Level B, or Level C.

C.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A, Level B, or Level C as specified (see 6.2).

5.1.1 Level A. The control unit or components shall be preserved by one of the following procedures depending upon procurement of complete control units or procurement of major components furnished separately.

5.1.1.1 Procurement of complete control units. Each control unit, without the batteries installed (see 5.1.1.1.1) shall be preserved in accordance with MIL-P-116, Method IIb. The inner container shall conform to PPP-B-636, Type CF, Class Domestic, Variety SW or DW, Grade 200. The control unit shall be cushioned and secured within the container to prevent movement or damage. The keys shall be placed in a cotton draw-string bag or heat-sealed bag and placed within the container. The outer container shall conform to PPP-B-636, W5c, style optional.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

5.1.1.1.1 Batteries. Each battery shall be separately preserved from the control unit in a close-fitting box conforming to PPP-B-636, W6c. The battery shall be cushioned and secured within the box to prevent free movement or damage. The box shall be closed and sealed as specified for Method V in the appendix to the box specification. The box shall be marked with the date of battery charging (see 3.6.1.4).

5.1.1.2 Procurement of major components furnished separately. In addition to the following requirements, major components furnished separately that are subject to electrostatic or electromagnetic damage shall be protected as specified in MIL-P-116 (see 5.3.2).

5.1.1.2.1 Batteries. Each battery shall be preserved as specified in 5.1.1.1.1.

5.1.1.2.2 Power supply. Power supplies shall be individually preserved in accordance with MIL-P-116, Method IIb. The inner container shall conform to PPP-B-636, Type CF, Class Domestic, Variety SW, Grade 175. The power supply shall be cushioned and secured within the container to prevent movement or damage. The outer container shall conform to PPP-B-636, W6c or W5c, style optional.

5.1.1.2.3 Other major components. Other major components shall be individually preserved in accordance with MIL-P-116, Method IIe, using a close-fitting box conforming to PPP-B-566 or PPP-B-676, Variety 1, style, type, and class optional.

5.1.1.2.3.1 Consolidation. Major components of like description, preserved as specified in 5.1.1.2.3, shall be consolidated together in a close-fitting box conforming to PPP-B-636, W6c or W5c, style optional, in quantities not to exceed the weight limitation of the box.

5.1.2 Level B. Preservation shall be as specified in 5.1.1 for Level A except Method 1A-14 shall be used in place of Method IIb and Method 1A-15 shall be used in place of Method IIe.

5.1.3 Level C. The control unit with the keys shall be preserved together. Batteries, power supplies, and any other major components furnished separately shall be individually preserved in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the first using activity.

5.2 Packing. Packing shall be Level A, Level B or Level C as specified (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:10Z
Check the source to verify that this is the current version before use.

5.2.1 Level A. The control unit or components preserved as specified in 5.1 shall be packed by one of the following procedures depending upon procurement of complete control units or procurement of major components furnished separately.

5.2.1.1 Procurement of complete control units.. Control units preserved as specified in 5.1 shall be packed together in a close-fitting box conforming to PPP-B-601, Overseas Type, Grade B, style optional, in quantities not to exceed the weight and size limitations of the box. Closure and strapping shall be in accordance with the Appendix to the box specification. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be Finish B. When specified (see 6.2) strapping shall be Finish A.

5.2.1.1.1 Batteries. Batteries preserved as specified in 5.1 shall be packed together in a close-fitting box conforming to PPP-B-601, Overseas Type, Grade B, style optional. Closure and strapping of the box shall be in accordance with the appendix to the box specification. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be Finish B. When specified (see 6.2) strapping shall be Finish A. The batteries shall be shipped separate from, but at the same time as the control unit.

5.2.1.2 Procurement of major components furnished separately. Components of like description, preserved as specified in 5.1, shall be packed together as specified in 5.2.1.1.

5.2.2 Level B. Control units without batteries, batteries, power supplies and major components furnished separately, preserved as specified in 5.1 shall be packed as specified in 5.2.1 as applicable, except boxes shall be Domestic Type. As an alternate, boxes conforming to PPP-B-636, V3c, V11c, V13c or V15c may be used when the quantity to be packed does not exceed the weight or size limitations of the box. Box closure and strapping shall be in accordance with the appendix of the applicable box specification.

5.2.3 Level C. Control units without batteries, batteries, power supplies and major components furnished separately, preserved as specified in 5.1, shall be packed in shipping containers that will afford adequate protection against damage during shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:10Z
Check the source to verify that this is the currentversion before use.

5.3 Marking. Marking for Level A, Level B or Level C shall be in accordance with MIL-STD-129 and the special marking specified in 5.3.1 and 5.3.2.

5.3.1 Special marking. Each shipping container shall be marked with project code labels in accordance with MIL-STD-129. The labels shall contain Project Code "JID". In addition, each shipping container containing the batteries shall be marked with arrows and word "up" to indicate stacking position.

5.3.2 Sensitive electronic components. Major components furnished separately, preserved as specified in 5.1.1.2 or 5.1.2 for components requiring electrostatic or electromagnetic protection, shall be additionally marked in accordance with MIL-STD-129 with a sensitive electronic device caution label and the following caution statement:

CAUTION

OBSERVE PRECAUTIONS FOR HANDLING

ELECTROSTATIC SENSITIVE ITEMS

6. NOTES

6.1 Intended use. The control unit, when used with the associated sensors, data transmitter, monitor cabinets and audible alarm is a component of the Joint-Services Interior Intrusion Detection System (J-SIIDS).

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When major components are to be furnished separately (see 3.1.1).
(c) Time frame required for submission of preproduction model, and number of control units and major components furnished separately required (see 3.2).
(d) When the Government will conduct any or all of the pre-production model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
(e) When initial production inspection is required and number of control units to be furnished (see 3.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

(f) When Government-loaned property is not required (see 3.10).
(g) Degree of preservation and degree of packing required (see 5.1 and 5.2).
(h) When other than Finish B strapping is required (see 5.2.1.1 and 5.2.1.1.1).

6.3 Preproduction model. Any changes or deviations of production control units and major components furnished separately, from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish control units and major components furnished separately conforming to this specification.

6.4 Definitions. The following definitions shall apply throughout this specification.

6.4.1 False alarm. An alarm generated by stimuli other than those for which the control unit is designed to alarm.

6.4.2 Alarm signals.

6.4.2.1 Intrusion alarm. Alarm received from the sensor subsystem in response to an intrusion or attempted intrusion of the control zone.

6.4.2.2 Tamper alarm. Alarm caused by an attack on the system itself (tamper or line supervisor alarms) and failures which cause an alarm.

6.4.2.3 Duress alarm. Alarm received from a latching alarm switch resulting from activation by security personnel.

6.4.3 Damage. Damage is defined as deformation, corrosion, loosening of parts, breakage, or change of fit of any component; physical change which impairs the mechanical integrity of the control unit; evidence of delamination or water penetration into integrated circuits, printed circuit boards, external components, or operational requirements resulting in noncompliance of the control unit with the provisions of this specification.

6.4.4 Quality assurance provisions (QAP). A QAP is a contractual requirement that supplements Section 4 of the specification. A QAP indicates the minimum requirements which must be inspected on the product drawings to verify the objectives of the product and assure interchangeability of the repair parts.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

6.5 Quality assurance provision (QAP) records. The contracting officer should require the contractor to maintain records of all QAP inspections. A suggested paragraph is as follows:

> "The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include, as a minimum, lot size, drawing requirements, actual measurements, number and type of deficiencies found, quantity approved, quantity rejected, and corrective action taken when applicable."

6.6 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each control unit.

6.7 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools, as necessary (including any special tools), and instructions regarding shipment of control units.

6.8 Government-loaned property. The contracting officer should arrange to loan the property specified in 3.10, when applicable (see 6.2).

6.9 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.4).

Custodian:

Army - ME

Air Force - 99

Preparing activity:

Army - ME

Project 6350-0056

Review activity:

DLA - GS

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

APPENDIX

TEST METHODS

FOR

CONTROL UNIT, ALARM SET

10. SCOPE

10.1 Scope. The test methods contained in this appendix are to determine whether control units and major components furnished separately, procured under this specification conform to the requirements specified. Figures A-1 through A-21 contain data for fabrication of test fixtures and preparation of test set-ups which are required to facilitate testing of the control units and major components furnished separately.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

TEST METHOD 1

FUNCTIONAL TEST

1. TESTS TO BE PERFORMED:

   (a) Power supply test.
   (b) Secure/access alarm test.
   (c) Secure mode test.
   (d) Access mode test.
   (e) Test/reset mode test.
   (f) Intrusion alarm delay test.
   (g) ALSA continuity test.
   (h) LAA inhibit test.
   (i) Battery voltage test.

2. TEST APPARATUS:

   (a) Digital voltmeter - Hewlett-Packard 34740A display with 34702A multimeter or equivalent.
   (b) Power supply (2) - Preston X-MOD-101 or equivalent.
   (c) Multimeter - Simpson Model 260 or equivalent
   (d) Resistor - 62 ohms, 10 watts.
   (e) Time interval meter - Eldorado 255 or equivalent.
   (f) Test fixture (see Figure A-1).
   (g) Test fixture schematic (see Figure A-2).
   (h) Control unit test set-up (see Figure A-3).
   (i) Control unit output conditions (see Table A1).

3. TEST PREPARATION:

   (a) Connect control unit to external test equipment as shown in Figure A-3.
   (b) Set all SENSOR SIMULATORS switches on test fixture to CLOSED and pull plunger out on door tamper switch.
   (c) Set control unit POWER switch on ON.
   (d) Place test fixture controls in the following positions:

Control	Position	Function
S1	1	MONITOR
S3	INST	TIME FUNCTION
S4	2	START TRIGGER
S5	1	STOP TRIGGER

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

Control	Position	Function
S6	OFF	LAA INHIBIT
S7	OFF	MANUAL
S10	OFF	DIALER

(e) Turn control unit mode switch to TEST/RESET and back to SECURE.

4. TEST PROCEDURE:

4.1 Primary power supply.

(a) Observing DVM, rotate MONITOR switch S1 from Position 1 thru Position 6. The DVM shall indicate 20 ±1 V at each position.
(b) Set control unit POWER switch to OFF.
(c) Remove fuse F2 and connect an external dc power supply to TB2-3 (+) and TB2-4 (ground).
(d) Adjust external power supply to 20 ±0.5 V.
(e) Set control unit POWER switch to ON.
(f) Turn control unit mode switch to TEST/RESET to establish normal operating conditions.
(g) Turn mode switch to SECURE. (Allow for exit time delay before proceeding.)
(h) Push RESET switch S8 on test fixture. All ALARMS indicator lights shall go out.
(i) Decrease external power supply voltage slowly to less than 19 V. ALARMS indicator lights INST, LATCH, and LAA shall illuminate.
(j) Readjust external power supply output to 20 V.
(k) Turn control unit mode switch to TEST/RESET and back to SECURE. (Allow for exit time delay before proceeding.)
(l) Push RESET switch S8.
(m) Increase external power supply voltage to greater than 21 V. ALARMS indicator lights INST, LATCH, and LAA shall illuminate.
(n) Readjust external supply output to 20 V.
(o) Set control unit mode switch to TEST/RESET.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

(p) Repeat steps (h), (i), (j), (l), (m), and (n) for the TEST/RESET position of the mode switch; except, ALARMS indicator light LAA shall not illuminate in this mode. However, the internal sonalert shall sound in Steps (i) and (m) and ALARMS indicator light A/S shall illuminate for all steps.

(q) Repeat Steps (h), (i), (j), (l), (m), and (n) for the ACCESS position of the mode switch. After Steps (j) and (n), turn the mode switch to TEST/RESET and back to ACCESS to reset latched alarms. Push RESET switch S8 to clear displays. ALARMS indicator light A/S shall illuminate for all steps.

(r) Set control unit POWER switch to OFF.

(s) Disconnect external power supply and replace fuse F2.

(t) Set control unit POWER switch to ON.

(u) Turn mode switch to TEST/RESET then back to SECURE. Wait 5 seconds for expiration of secure/access delay.

(v) Push RESET switch S8. All ALARMS indicator lights shall go out.

(w) Set TIMER FUNCTION switch S3 to AC FAIL, STOP TRIGGER switch S5 to 2, and reset time interval meter.

(x) Simultaneously unplug ac power from the control unit and set MANUAL switch S7 to ON. The timer interval meter shall begin running. After 2.5 to 5 seconds (3 seconds normal), the timer shall stop and ALARMS indicator light AC FAIL shall illuminate.

(y) Turn off MANUAL switch S7.

(z) Reapply ac power to the control unit.

(aa) Push RESET switch S8, ALARMS indicator light AC FAIL shall go out.

4.2 Secure/access alarm test.

(a) Set control unit mode switch to SECURE. Wait approximately 6 seconds and push RESET switch S8. Set TIMER FUNCTION switch S3 to S/A, START TRIGGER switch S4 to 1, and STOP TRIGGER switch S5 to 2.

(b) All ALARMS indicator lights shall be off. Reset time interval meter to zero.

(c) Turn control unit mode switch to TEST/RESET and back to SECURE. ALARMS indicator light S/A shall illuminate and the time interval meter shall begin running. When the timer stops, the meter shall indicate 4.5 to 6 seconds.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

(d) Push RESET switch S8. ALARMS indicator light S/A shall go out.
(e) Rapidly turn mode switch to ACCESS and back to SECURE. The conditions of Step (c) shall occur.
(f) Push RESET switch S8. ALARMS indicator light S/A shall go out.

4.3 Secure mode test.

(a) Set TIMER FUNCTION switch S3 to INST, START TRIGGER switch S4 to 2, and STOP TRIGGER switch S5 to 1.
(b) Set LATCH/NONLATCH switch on status processor to LATCH.
(c) Push RESET switch S8 and reset time interval meter.
(d) Momentarily set SENSOR SIMULATORS switch A1 to OPEN to simulate an intrusion alarm. ALARMS indicator light INST shall illuminate, the proper LED on board A12 shall illuminate, and the time interval meter shall begin running.
(e) When the timer stops, ALARMS indicator lights LATCH and LAA shall illuminate, and the meter shall indicate 10 $\pm$2 seconds.
(f) Turn control unit mode switch to TEST/RESET and back to SECURE. Wait 6 seconds.
(g) Push RESET switch S8. All ALARMS indicator lights shall go out.
(h) Repeat Steps (c) thru (g) for SENSOR SIMULATORS switches B1, C1, D1, and E1.
(i) Set LATCH/NONLATCH switch to NONLATCH.
(j) Set TIMER FUNCTION switch S3 to LATCH/NL, START TRIGGER switch S4 to 2, and STOP TRIGGER switch S5 to 1.
(k) Repeat Steps (c) thru (i); except, in Step (e) when the timer stops, push RESET switch S8. ALARMS indicator light LATCH shall go out.
(l) Set LATCH/NONLATCH switch to LATCH.
(m) Set TIMER FUNCTION switch S3 to INST, START TRIGGER switch S4 to 2, and STOP TRIGGER switch S5 to 1.
(n) Turn mode switch to TEST/RESET and back to SECURE. (Allow for exit time delay before proceeding.)
(o) Push RESET switch S8.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:10Z
Check the source to verify that this is the current version before use.

(p) Momentarily set SENSOR SIMULATORS switch A3 to OPEN to simulate a tamper alarm. ALARMS indicator lights INST, LATCH, and LAA shall illuminate; and the time interval meter shall begin running.
(q) When the timer stops, it shall indicate 10 ±2 seconds.
(r) Push RESET switch S8. ALL ALARMS indicator lights shall go out. The LATCH and LAA indicator lights shall illuminate when the switch is released.
(s) Turn mode switch to TEST/RESET and back to SECURE. (Allow for exit time delay before proceeding.)
(t) Push RESET switch S8. ALL ALARMS indicator lights shall go out.
(u) Repeat Steps (p) thru (t) using SENSOR SIMULATORS switches B3, C3, D3, E3, and DIALER tamper switch S10.
(v) Repeat Steps (p) thru (u) with the LATCH/NONLATCH switch in NONLATCH position; except in Step (r) when S8 is pushed, ALARMS indicator lights INST and LATCH only will go out and LAA will remain on.
(w) Set LATCH/NONLATCH switch to LATCH.
(x) Turn mode switch to TEST/RESET and back to SECURE. (Allow for exit time delay before proceeding.)
(y) Push RESET switch S8 and reset time interval meter.
(z) Momentarily set SENSOR SIMULATORS switch F1 to OPEN to simulate a duress alarm. ALARMS indicator lights INST and LATCH shall illuminate, the time interval meter shall begin running, and the proper LED on A12 shall illuminate.
(aa) When the timer stops, it shall indicate 10 ±2 seconds.
(bb) Set LATCH/NONLATCH switch to NONLATCH.
(cc) Repeat Steps (x) thru (aa) except in Step (aa) when timer stops, push RESET switch S8. ALARMS indicator lights LATCH and INST shall go out.
(dd) Turn mode switch to TEST/RESET and back to SECURE. (Allow for exit time delay before proceeding.)
(ee) Set LATCH/NONLATCH switch to LATCH.
(ff) Push RESET switch S8, and reset time interval meter.
(gg) Open control unit tamper switch momentarily. ALARMS indicator lights INST, LATCH, and LAA shall illuminate; and the time interval meter shall begin running.
(hh) When timer stops, the meter shall indicate 10 ±2 seconds.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:10Z
Check the source to verify that this is the current version before use.

(ii) Turn mode switch to TEST/RESET and back to SECURE. (Allow for exit time before proceeding.)
(jj) Push RESET switch S8 and reset time interval meter.
(kk) Repeat Steps (gg) thru (jj) for the NONLATCH position; except following Step (hh), push RESET switch S8. ALARMS indicator lights INST and LATCH shall go out.
(ll) Monentarily set SENSOR SIMULATORS A1 to open, wait 1 second then switch B1 to open. LED A on board A12 shall illuminate and LED B shall not illuminate.
(mm) Repeat Step (ii).
(nn) Momentarily set SENSOR SIMULATORS switch A1 to OPEN. Wait 500 ms then set switch B1 to OPEN. LED's A and B shall illuminate.
(oo) Repeat Step (ii).
(pp) Push TRANSMITTER RESYNC switch on status processor. The test fixture RESYNC indicator light shall illuminate.

4.4 Access mode test.

(a) Repeat the tests of paragraph (4.3) with the mode switch at ACCESS instead of SECURE. (See Table A-1 for differences in outputs from the secure mode.)
(b) Turn DVM switch S2 to 2.
(c) Turn MONITOR switch S1 from Position 1 through 6. The digital voltmeter shall indicate 24 to 28.5 V at each position except Position 6.
(d) Set TIMER FUNCTION switch S3 to EGRESS, START TRIGGER switch S4 to 2, and STOP TRIGGER switch S5 to 2.
(e) Push RESET switch S8 and reset time interval meter.
(f) Turn potentiometer A9R6 fully clockwise.
(g) Turn mode switch from ACCESS to SECURE.
(h) Set SENSOR SIMULATORS switch A1 to OPEN. ALARMS indicator light INST shall illuminate, the LATCH and LAA indicator lights shall be off, and the time interval meter shall begin running.
(i) When the timer stops, ALARMS indicator lights LATCH and LAA shall illuminate, indicator light INST shall remain on, and the time interval meter shall indicate 20 $\pm$4 seconds.
(j) Turn potentiometer A9R6 approximately 120 degrees counterclockwise.
(k) Set SENSOR SIMULATORS switch A1 to CLOSED.
(l) Turn mode switch to ACCESS.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

(m) Push RESET switch S8 and reset time interval meter.
(n) Turn mode switch to SECURE.
(o) Repeat Steps (h) and (i); except in Step (i), the meter shall indicate 100 ±12 seconds.
(p) Set SENSOR SIMULATORS switch A1 to CLOSED.
(q) Turn mode switch to TEST/RESET.
(r) Push RESET switch S8 and reset time interval meter.
(s) Turn potentiometer A9R6 fully clockwise.

4.5 Test/reset mode test.

(a) Repeat the tests of paragraph 4.3 in TEST/RESET mode. (See Table A-1 for differences in output conditions from secure mode.)

4.6 Intrusion alarm delay test.

(a) Set mode switch to SECURE. (Allow for exit time delay before proceeding.)
(b) Push RESET switch S8 and reset time interval meter.
(c) Turn potentiometer A10R8 fully clockwise.
(d) Set TIMER FUNCTION switch S3 to ACCESS, START TRIGGER Switch S4 to 2, and STOP TRIGGER switch S5 to 2.
(e) After the exit time delay has elapsed, simulate an intrusion by setting any SENSOR SIMULATORS load switch A1 through E1 to OPEN. ALARMS indicator light INST shall illuminate and indicator lights LATCH and LAA shall be off. The time interval meter shall begin running.
(f) When the timer stops, ALARMS indicator lights LATCH and LAA shall illuminate, indicator light INST shall remain on, and the time interval meter shall indicate 10 ±2 seconds.
(g) Set SENSOR SIMULATORS switch which was opened in Step (e) to CLOSED.
(h) Turn mode switch to TEST/RESET then back to SECURE. (Allow for exit time before proceeding.)
(i) Turn potentiometer A10R8 approximately 120 degrees counterclockwise, or as required for 90 seconds.
(j) Repeat Steps (e) and (f); except in Step (f), the time interval meter shall indicate 90 ±10 seconds.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

4.7 ALSA continuity.

(a) Using an ohmmeter, check continuity between the following points:

Control unit	ALSA connector
F1	B
F3	A
A1	L
A3	S and K
A4	M
A6	U
B1	R
B3	P
C1	J
C3	H
D1	F
D3	E
E1	D
E3	C

4.8 LAA inhibit test.

(a) Turn mode switch to TEST/RESET then to SECURE. (Allow for exit time delay before proceeding.)
(b) Push RESET switch S8 and reset time interval meter.
(c) Set LAA INHIBIT switch S6 to ON.
(d) Set SENSOR SIMULATORS switch F3 to OPEN to simulate a tamper alarm. ALARMS indicator light INST shall illuminate and indicator lights LATCH and LAA shall be off.
(e) Turn off LAA INHIBIT switch S6. ALARMS indicator light INST shall remain on and indicator lights LATCH and LAA shall illuminate.
(f) Set SENSOR SIMULATORS switch F3 to CLOSED.
(g) Turn mode switch to TEST/RESET.

4.9 Battery voltage test.

(a) Connect a 62-ohm, 10-watt resistor from Terminal TB2-4 to TB2-1 to simulate a sensor load of 325 mA.
(b) Turn mode switch to SECURE.
(c) Using the digital voltmeter, measure the battery terminal voltage. It shall be 24 V minimum.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

TEST METHOD 2

STANDBY POWER TEST

1. TEST APPARATUS:

(a) Digital voltmeter - Hewlett-Packard 34740A display with 34702A multimeter or equivalent.
(b) Power supply - Preston X-MOD-101 or equivalent.
(c) Multimeter - Simpson Model 260 or equivalent.
(d) Resistor - 62 ohms, 10 watts.
(e) Resistor - 96 ohms, 10 watts.
(f) Test fixture (see Figure A-1).
(g) Test fixture schematic (see Figure A-2).
(h) Control unit test set-up (see Figure A-3).

2. TEST PREPARATION:

(a) Connect control unit to external test equipment as shown in Figure A-3.
(b) Set all SENSOR SIMULATORS switches on the test fixture to CLOSED and close the tamper switch by pulling plunger out.
(c) Set control unit POWER switch to ON.
(d) Place test fixture controls in the following positions:

Control	Position	Function
S1	1	MONITOR
S2	1	DVM
S3	INST	TIMER FUNCTION
S4	2	STARTER TRIGGER
S5	1	STOP TRIGGER
S6	OFF	LAA INHIBIT
S7	OFF	MANUAL
S10	OFF	DIALER

(e) Turn mode switch to the TEST/RESET position to initially establish normal operating conditions.
(f) Turn mode switch to SECURE position.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

3. TEST PROCEDURE:

(a) Observing DVM, rotate MONITOR switch S1 from Position 1 thru Position 6. The DVM shall indicate 20 ±1 V at each position.
(b) Set control unit POWER switch to OFF.
(c) Connect an external load between terminals A-4 and A-5 to simulate sensor load of 325 mA (62 ohms, 10 watts).
(d) Connect an external load between terminals A-7 and A-5 to simulate external access command load of 250 mA (96 ohms, 10 watts).
(e) Connect an ammeter in series with positive battery terminal and power supply.
(f) Set POWER switch to ON and allow the battery to fully charge.
(g) After the battery is fully charged as indicated on ammeter by a charge current of 100 mA or less, remove ac power by unplugging the ac line cord.
(h) Turn mode switch momentarily to TEST/RESET position and back to SECURE.
(i) Allow control unit to operate on battery power for 24 hours.
(j) At the end of the 24-hour period, perform SECURE/ACCESS alarm test, Test Method 1 (4.2); and secure mode test, Test Method 1 (4.3), except omit Steps (h) and (u).

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:10Z
Check the source to verify that this is the current version before use.

TEST METHOD 3

TAMPER SWITCH TOLERANCE TEST

1. TEST APPARATUS:

   (a) Six-inch ruler.
   (b) Control unit test set-up (see Figure A-3).

2. TEST PREPARATION:

   (a) Connect control unit to external test equipment as shown in Figure A-3.
   (b) Set all SENSOR SIMULATORS switches on test fixture to CLOSED.
   (c) Set control unit switch to ON.
   (d) Close and lock door.
   (e) Set mode switch to TEST/RESET.

3. TEST PROCEDURE:

   (a) Note the door flange position on the right side of the cabinet near the door lock. Make a small mark on the cabinet side panel to record the exact position of the door flange.
   (b) Unlock and slowly open the cabinet door. Mark the door flange position on the side panel at its exact position when the internal alarm sounds. The door flange shall not open beyond the edge of the side panel before the internal buzzer sounds.
   (c) Measure between marks. The measurement shall be not less than 1/8 inch nor more than 1/4 inch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

TEST METHOD 4

MODE SWITCH LIFE TEST

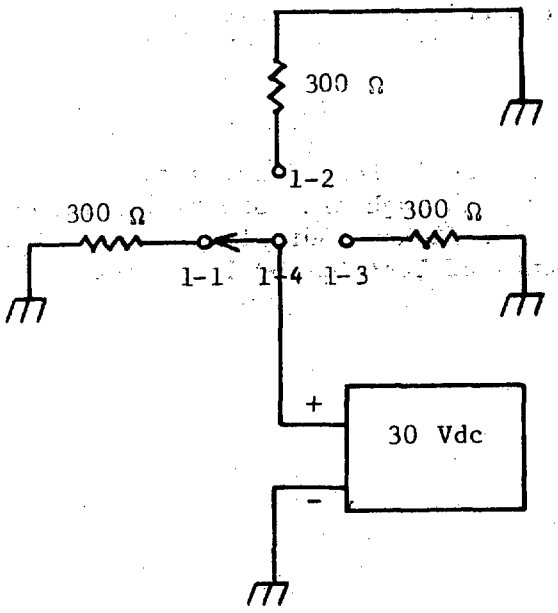
1. TEST APPARATUS:

   (a) Power supply (3) - Preston Model X-MOD-101 or equivalent.
   (b) Resistor (9) - 300 ohms.
   (c) Mode switch actuating device - As required.

2. TEST PROCEDURE:

   (a) The following test shall be performed prior to installation of the switch in the control unit:

      (1) Connect a 30 V, 100mA load across switch terminals as shown below (Section 1 of switch only shown (all sections to be tested).
      (2) Actuate switch 50,000 times.
      (3) After completion of life test, the mode switch shall show no visual sign of damage and shall function as specified after installation into the control unit.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

TEST METHOD 5

BURN-IN

1. TEST APPARATUS:

   (a) Digital voltmeter - Hewlett-Packard 34740A display with 34702A multimeter or equivalent.
   (b) Power supply (2) - Preston X-MOD-101 or equivalent.
   (c) Multimeter - Simpson Model 260 or equivalent.
   (d) Resistor - 62 ohms, 10 watts.
   (e) Time interval meter - Eldorado 255 or equivalent.
   (f) Test fixture (see Figure A-1).
   (g) Test fixture schematic (see Figure A-2).
   (h) Control unit test set-up (see Figure A-3).

2. TEST PREPARATION:

   (a) Connect control unit to external test equipment as shown in Figure A-3.
   (b) Set all SENSOR SIMULATORS switches on test fixture to CLOSED and pull plunger out on door tamper switch.
   (c) Set LAA INHIBIT, MANUAL, and DIALER switches S6, S7, and S10 on test fixture to the OFF position.
   (d) Set control unit POWER switch to ON.
   (e) Turn control unit mode switch to TEST/RESET then to SECURE.

3. TEST PROCEDURE:

   (a) Nine times each 24 hours during the 72 hours, exercise the control unit through each of its functions.
   (b) Following the 72-hour burn-in the control unit shall meet the requirements of Test Method 1, Functional Test.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:10Z
Check the source to verify that this is the current version before use.

TEST METHOD 6

POWER SUPPLY ASSEMBLY

1. TEST APPARATUS:

(a) Oscilloscopes (Scope 1, Scope 2), - Tektronic, Model 5440 with Type 5A48 plug in or equivalent.
(b) Voltmeters dc, (M3, M5, M7, M9, M13) - Weston, Model 911 dc voltmeter or equivalent.
(c) Ammeters dc, (M4, M6, M8) - Weston, Model 911 dc ammeter or equivalent.
(d) Voltmeter, ac (M1, M11, M12) - Weston, Model 912 ac voltmeter or equivalent.
(e) Ammeter ac, (M10) - Weston, Model 912 ac ammeter or equivalent.
(f) Multimeter (M14) - Simpson, Model 260 or equivalent.
(g) Autotransformer (VR1) - General Radio, Model W20M or equivalent.
(h) Power supply (PS1) - Hewlett-Packard, Model 6289A or equivalent.
(i) Battery (B1) - Eagle-Picher, Model CF24V18, no substitute.
(j) SPST switches (S1-S9) - Leviton, Model 19916 or equivalent.
(k) Current probe (M2) - Hewlett-Packard, Model 428 or equivalent.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

(l) Resistor, 2k ohm - Allen-Bradley, Type GB, 2k ohm ±5%, 1 watt or equivalent.
(m) Resistor, 66 ohm - Ohmite/Dividohm, 75 ohm, adjustable 12 watts, P/N 1011 or equivalent
(n) Resistor, 33 ohm - Ohmite/Dividohm, 50 ohm, adjustable, 25 watts, P/N 0366 or equivalent.
(o) Resistor, 5.5 ohm - Ohmite/Dividohm, 10 ohm, adjustadjust- 225 watts, P/N 1357 or equivalent
(p) Resistors, (112 ohm, 100 ohm) - Ohmite/Dividohm, 150 ohm, adjustable, 12 watts, P/N 1013 or equivalent
(q) Resistor, 31.5 ohm - Ohmite/Dividohm, 50 ohm, adjustable, 12 watts, P/N 1010 or equivalent.
(r) Sound level meter - B & K, Type 2209 or equivalent.
(s) Test fixture (see Figure A-4).

2. TEST PROCEDURE:

2.1 General. Unless otherwise specified, all testing will be conducted at prevailing room conditions.

2.2 Burn-in. Burn-in will be 72 hours minimum and will be conducted under the following conditions:

(a) Close S1, S2, S4, S5, S7, and S8 and open S3, S6, and S9, place the power switch in the ON position.
(b) Input voltage will be 115 Vac at 60 Hz (TB-1 to TB-2) as indicated on M1.
(c) The 20 Vdc output will be at 0.600 ampere load (TB2-3 to TB2-4) as indicated on M6.
(d) The 27 Vdc battery charger output will be loaded with one CF24V18 battery (TB2-1 to TB2-4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

(e) The 6.3 Vac output will be at a 0.200 ampere load (TB1-5 to TB1-6) as indicated on M10.
(f) The 5 Vdc output will be at 50 mA load (TB2-5 to TB2-4) as indicated on M8.
(g) The 27 Vdc output will be at 0.250 ampere load (TB2-2 to TB2-4) as indicated on M4.
(h) During the burn-in period, the 115 Vac input power will be removed by opening S1 at the completion of 20 hours of operation and the power supply assembly will be powered by the standby battery for a period of 4 hours. At the completion of 24 hours of operation, the 115 Vac input will be restored by closing S1. This procedure will be repeated not less than 3 times over the 72 hours burn-in period.
(i) The burn-in is to be completed prior to the remaining test procedures.
(j) Any failure of any component will constitute a failure.

2.3 Output voltage levels.

(a) Connect the test apparatus as shown in Figure A-4. Set power supply assembly power switch in the OFF position and apply 115 Vac, 60 Hz line power to the autotransformer (VRI).
(b) Open S1 through S9.
(c) Adjust the VRI for 115 Vac on M1.
(d) Close S1.
(e) Place the power supply assembly power switch in the ON position.
(f) The following output voltage levels shall be observed:

Output	Location	Meter	Voltage
1	TB2-3 to TB2-4	M7	20 Vdc ±1 V
2	TB2-1 to TB2-4	M3	27 Vdc ±0.5 V at 25° C
3	TB2-5 to TB2-4	M9	5 Vdc ±0.25 V
4	TB1-5 to TB1-6	M11	6.3 Vac ±1 V
5	TB2-2 to TB2-4	M5	27 Vdc ±0.5 V

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:10Z
Check the source to verify that this is the current version before use.

(g) The output voltages must be within the above levels to be acceptable.

2.4 Line regulation.

(a) Close S1, S4, S5, S7, and S8 and open S2, S3, S6, and S9.
(b) Place the power supply assembly power switch in the ON position.
(c) Observe the following output current levels:

Output	Location	Meter	Current
1	TB2-3 to TB2-4	M6	0.600 A
2	TB2-1 to TB2-4	M2	0 A
3	TB2-5 to TB2-4	M8	0.050 A
4	TB1-5 to TB1-6	M10	0.200 A
5	TB2-2 to TB2-4	M4	0.250 A

(d) Adjust the VRI for a 115 Vac reading on M1.
(e) Observe and record the following output voltage levels:

Output	Location	Meter
1	TB2-3 to TB2-4	M7
2	TB2-1 to TB2-4	M3
3	TB2-5 to TB2-4	M9
4	TB1-5 to TB1-6	M11
5	TB2-2 to TB2-4	M5

(f) Adjust the VRI for 125 Vac.
(g) Repeat Step (e).
(h) Adjust the VRI for 110 Vac.
(i) Repeat Step (e).
(j) For acceptable line regulation, the output voltage measurements taken in Steps (g) and (i) shall not differ from those taken in Step (e) by more than the following when measured at a constant temperature:

(1) Output 1 $\pm$0.5%
(2) Output 2 $\pm$1%
(3) Output 3 $\pm$1%
(4) Output 4 $\pm$10%
(5) Output 5 $\pm$1%

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

2.5 Load Regulation.

(a) Close S1 and S2 and open S3, S4, S5, S6, S7, S8, and S9.
(b) Adjust VR1 for 115 Vac as indicated on M1.
(c) Observe and record the output voltages as follows:

Output	Location	Meter
1	TB2-3 to TB2-4	M7
3	TB2-5 to TB2-4	M9
4	TB1-5 to TB1-6	M11
5	TB2-2 to TB2-4	M5

(d) Close S4, S5, S7, and S8. Repeat the voltage readings as in Step (c). M6 shall indicate 0.600 A, M8 shall indicate 0.050 A, M10 shall indicate 0.200 A and M4 shall indicate 0.250 A.
(e) Open S4, S5, S7, and S8 and close S6. Repeat the voltage readings as in Step (c). M6 shall indicate 0.300 A and M8, M10, and M4 shall indicate 0 A.
(f) For acceptable load regulation, the output voltage measurements taken in Steps (d) and (e) shall not differ from those taken in Step (c) by more than the following:

(1) Output 1 ±1%
(2) Output 3 ±1%
(3) Output 4 ±5%
(4) Output 5 ±1%

2.6 Ripple and noise (periodic and random deviations).

(a) Open S3, S6, and S9 and close S1, S2, S4, S5, S7, and S8.
(b) Measure the output ac peak-to-peak voltage at Output 1 (TB2-3 to TB2-4) with Scope 1. The lead lengths from the scope to TB2 should be as short as possible.
(c) Measure the output ac peak-to-peak voltage at Output 3 (TB2-5 to TB2-4) with Scope 2. The lead lengths from the scope to the TB2 should be as short as possible.
(d) For acceptable ripple content the ac peak-to-peak voltage shall be less than 300 mv for Output 1 and 200 mv for Output 3.

Source: https://assist.dla.mil -- Downloaded: 2026l0-26T04:10Z
Check the source to verify that this is the current version before use.

2.7 Standby power.

Battery operational and charging characteristics.

(a) Open S3, S4, S5, S6, S7, S8, and S9 and close S1 and S2.
(b) Place the power switch in the ON position and adjust VR1 for 115 Vac as indicated on M1.
(c) The power supply is to operate on ac power until the battery is fully charged. M3 will indicate 27 Vdc ±0.5 volts and M2 will indicate less than 250 mA when the battery is fully charged. Observe polarity for M2.
(d) Close S4, S5, S7, and S8 and open S1. Observe polarity for M2.
(e) M7 shall indicate 20 ±1 Vdc, M5 shall indicate 27 ±0.5 Vdc, M9 shall indicate 5 ±0.25 Vdc, and M11 shall indicate 0 Vac.
(f) Close S3 and allow the power supply assembly to operate from standby battery power until M3 indicates 24 ±0.5 Vdc.
(g) Open S3 and close S-1. Observe polarity for M2.
(h) M3 shall indicate 26 to 30 Vdc until the battery is fully charged. When the battery is fully charged, M3 will indicate 27 ±0.5 Vdc and M2 will indicate less than 250 mA.
(i) Reduce the input voltage to 110 Vac as indicated on M1. The current on M2 shall indicate less than 250 mA.
(j) For acceptance the battery operational and charging characteristics shall be within the above specified limits.

2.8 Auxiliary outputs and inputs

2.8.1 110 - 125 Vac, 48 - 62 Hz, single phase output.

(a) Open S3, S4, S5, S6, S7, S8, and S9 and close S1 and S2.
(b) Place the power switch in the ON position and adjust VR1 for 115 Vac as indicated on M1.
(c) M12 shall indicate 115 ±1 Vac.
(d) Adjust VR1 to 110 Vac as indicated on M1.
(e) M12 shall indicate 110 ±1 Vac.
(f) Adjust VR1 to 125 Vac as indicated on M1.
(g) M12 shall indicate 125 ±1 Vac.
(h) For acceptance the 110 - 125 Vac output characteristics shall be within the above specified limits.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

2.8.2 4 - 6 Vdc output.

(a) Open S3, S4, S5, S6, S7, S8, and S9 and close S1 and S2.
(b) Place the power switch in the ON position and adjust VR1 for 115 Vac as indicated on M1.
(c) M13 shall indicate 5 ±1 Vdc.
(d) For acceptance the 4 - 6 Vdc output characteristics shall be within the above specified limits.

2.8.3 Output, ac presence.

(a) Open S3, S4, S5, S6, S7, S8, and S9 and close S1 and S2.
(b) Place the power switch in the ON position and adjust VR1 for 115 Vac as indicated on M1.
(c) M14 shall indicate 1 ±1 ohm.
(d) Disconnect the input power to VR1.
(e) M14 shall indicate 200 ±20k ohms.
(f) For acceptance the ac presence output characteristics shall be within the above specified limits.

2.8.4 Sonalert input.

(a) Open S1, S2, S3, S4, S5, S6, S7, and S8 and close S9.
(b) Adjust PS1 for 20 ±1 Vdc.
(c) The sonalert shall produce a minimum audible sound of 60 dB at a distance of 10 feet as measured with the sound level meter.
(d) Open S9, the audible sound shall stop.
(e) For acceptance the sonalert input characteristics shall be within the above specified limits.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

TEST METHOD 7

TAMPER/ALARM PRINTED WIRING BOARD ASSEMBLY
(13220E4213)

A1 THROUGH A6 CARDS

FUNCTIONAL TEST

1. TEST APPARATUS:

(a) Power supply, Hewlett-Packard 6202B or equivalent.
(b) Digital voltmeter, Fairchild 7000A or equivalent.
(c) Test fixture schematic (see Figure A-5).
(d) Test fixture schematic (see Figure A-6).
(e) A-1 - A-6 and A-7 card test set-up (see Figure A-7).

2. TEST PROCEDURE:

(a) Connect test apparatus as shown in Figure A-7.

NOTE: Test fixture shall be kept at room ambient temperature, with no more than a $\pm 5^{\circ}$ F swing during tests, without recalibrating the fixture. When performing functional test as a part of the low or high temperature test, a connecting cable must be run from the test fixture (outside the temperature chamber) to the circuit card (inside the chamber). Cable length should be less than two (2) feet, if possible.

(b) Turn off power supply.
(c) Install card to be tested in test fixture and connect test point adapter.
(d) Turn on power supply, and adjust output to 19.9 ± 0.1 Vdc using the DVM.
(e) Place Switch 1 in Position 1 (Intrusion).
(f) Momentarily activate Switch 2 (65 ms timer) and observe lights.

PASS - Amber and green lights appear simultaneously for approximately 2 seconds.

FAIL - Amber and red lights appear simultaneously for approximately 2 seconds.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

(g) Momentarily activate Switch 3 (135 ms timer) and observe lights.

PASS - See Step (f)

FAIL - See Step (f).

NOTE: If a failure occurs during Step (g), repeat Step (g) two more times. Two failures out of three attempts constitutes a test failure and rejection of the card.

(h) Place Switch 1 in Position 2 (Tamper).
(i) Repeat Step (f).
(j) Repeat Step (g).
(k) Turn off power supply.
(l) Remove test point adapter from card, and remove card from test fixture.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:10Z
Check the source to verify that this is the currentversion before use.

TEST METHOD 8

VOLTAGE MONITOR PRINTED WIRING BOARD ASSEMBLY
(13220E4214)

A7 CARD

FUNCTIONAL TEST

1. TEST APPARATUS:

(a) Power supply, Hewlett-Packard 6202B or equivalent.
(b) Digital voltmeter, Fairchild 7000A or equivalent.
(c) Test fixture (see Figure A-5).
(d) Test dixture schemetic (see Figure A-6).
(e) A-1 - A-6 and A-7 card test set-up (see Figure A-7).

2. TEST PROCEDURE:

(a) Connect test apparatus as shown in Figure A-7.

NOTE: Test fixture shall be kept at room ambient temperature, with no more than a $\pm 5^{\circ}$ F swing during tests, without recalibrating the fixture. When performing functional test as a part of the low or high temperature test, a connecting cable must be run from the test fixture (outside the temperature chamber) to the circuit card (inside the chamber). Cable length should be less than two (2) feet, if possible.

(b) Turn off power supply.
(c) Install A7 card to be tested in test fixture, and connect test point adapter.
(d) Turn on power supply, adjust voltage to 19 to 19.15 Vdc using the DVM, and observe test fixture lights.

PASS - White light continuously ON.

FAIL - White light intermittent or continuously OFF.

(e) Adjust power supply voltage to 20.85 to 21 Vdc using the DVM, and observe test fixture lights.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

PASS - White light continuously ON.

FAIL - White light intermittent or continuously OFF.

(f) Adjust power supply voltage to 18.5 ±0.1 Vdc using the DVM and observe test fixture lights.

PASS - White light continuously OFF.

FAIL - White light intermittent or continuously ON.

(g) Adjust power supply to 21.5 ±0.1 Vdc using the DVM and observe test fixture lights.

PASS - White light continuously OFF.

FAIL - White light intermittent or continuously ON.

(h) Adjust power supply output to 19.9 ±0.1 Vdc using the DVM.
(i) Place Switch 1 in Position 1 (Door tamper).
(j) Momentarily activate Switch 2 (65 ms timer) and observe lights.

PASS - Amber and green lights appear simultaneously for approximately 2 seconds.

FAIL - Amber and red lights appear simultaneously for approximately 2 seconds.

(k) Momentarily activate Switch 3 (135 ms timer) and observe lights.

PASS - See Step (j).

FAIL - See Step (j).

NOTE: If a failure occurs during Step (k), repeat Step (k) two more times. Two failures out of three attempts constitutes a test failure and rejection of the card.

(l) Place Switch 1 in Position 2 (Telephone dialer tamper).
(m) Repeat Step (j).

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:10Z
Check the source to verify that this is the currentversion before use.

(n) Repeat Step (k).
(o) Place Switch 4 in Position 1 (Alarm), momentarily activate S5, and observe test fixture lights.

PASS - White light will turn OFF, and green light will turn ON while S4 is activated. When S4 is released, white light will turn ON and green light will turn OFF.

FAIL - Any combination of white and green lights other than specified under the PASS condition or the activation of the red light.

(p) Place Switch 4 in Position 2 (Nonalarm), momentarily activate S6, and observe test fixture lights.

PASS - White light will be ON and green light will turn ON while S4 is activated. When S4 is released, white light will be ON, and green light will turn OFF.

FAIL - Any combination of white and green lights other than specified under PASS condition or the activation of the red light.

(q) Turn off power supply.
(r) Remove test point adapter from A7 card, and remove card from test fixture.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

TEST METHOD 9

AUDIBLE ALARM PRINTED WIRING BOARD ASSEMBLY
(13220E4215)

A8 CARD

FUNCTIONAL TEST

1. TEST APPARATUS:

   (a) Power supply, Hewlett-Packard 6202B or equivalent (two required).
   (b) Digital voltmeter, Fairchild 7000A or equivalent (two required).
   (c) Test fixture (see Figure A-8).
   (d) Test fixture schematic (see Figure A-9).
   (e) A8 card test set-up (see Figure A-10).

2. TEST PROCEDURE:

   (a) Connect test apparatus as shown in Figure A-10.

   NOTE: Test fixture shall be kept at room ambient temperature, with no more than a $\pm 5^{\circ}$ F swing during tests, without recalibrating the fixture. When performing functional test as a part of the low or high temperature test, a connecting cable must be run from the test fixture (outside the temperature chamber) to the circuit card (inside the chamber). Cable length should be less than two (2) feet, if possible.

   (b) Turn off both power supplies.
   (c) Install A8 card in test set.
   (d) Turn on both the 20 Vdc and 5 Vdc power supplies, and adjust to 19.9 $\pm$0.1 Vdc and 4.9 $\pm$0.1 Vdc, respectively, using the DVM's.
   (e) Place S1, S4, S5, and S6 in Position 1, and momentarily depress reset switch S3.

   PASS - All test indicators are off.

   FAIL - Any test set indicator turned ON.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

NOTE: If the yellow test-in-progress indicator is ON, wait until it turns OFF before depressing S3.

(f) Wait to allow the A8 card circuitry to stabilize - approximately 30 seconds.
(g) Verify that the yellow test-in-progress indicator is OFF.
(h) Momentarily depress S2, and observe test set low and high limit indicators.

PASS - Yellow test-in-progress indicator will light, and before it extinguishes both the low and high limit green indicators will light and will remain lit when the test in progress indicator turns off.

FAIL - Either or both low or high limit red indicators light.

(i) Wait until the test-in-progress indicator goes OFF; place S4 in Position 2; and momentarily depress reset switch S3.
(j) Repeat Steps (g) and (h).
(k) Place S1 and S5 in Position 2, and momentarily depress reset switch S3.
(l) Repeat Steps (g) and (h).
(m) Wait until the test in progress indicator goes OFF; then depress S7, and observe the LAA LED.

PASS - The LED is OFF before depressing the switch; the LED is ON while the switch is depressed; and the LED turns OFF when the switch is released.

FAIL - The LED is ON before depressing the switch; the LED remains OFF while the switch is depressed, or the LED remains ON when the switch is released.

(n) Place S6 in Position 2; depress S8, and observe the LAA LED.

PASS - See Step (m).

FAIL - See Step (m).

(o) Depress S9, and observe the ACC LED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

PASS - See Step (m).

FAIL - See Step (m).

(p) Depress S10, and observe the SOD LED.

PASS - See Step (m).

FAIL - See Step (m).

(q) Turn off both power supplies.
(r) Remove the A8 card from the test set.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:10Z
Check the source to verify that this is the currentversion before use.

TEST METHOD 10

INSTANTANEOUS ALARM PRINTED WIRING BOARD ASSEMBLY
(13220E4216)

A9 CARD

FUNCTIONAL TEST

1. TEST APPARATUS:

(a) Power supply, Hewlett-Packard 6202B or equivalent (two required).
(b) Digital voltmeter, Fairchild 7000A or equivalent (two required).
(c) Test fixture (see Figure A-11).
(d) Test fixture schematic A (see Figure A-12).
(e) Test fixture schematic B (see Figure A-13).
(f) A9 card test set-up (see Figure A-14).

2. TEST PROCEDURE:

(a) Connect test apparatus as shown in Figure A-14.

NOTE: Test fixture shall be kept at room ambient temperature, with no more than a $\pm 5°$ F swing during tests, without recalibrating the fixture. When performing functional test as a part of the low or high temperature test, a connecting cable must be run from the test fixture (outside the temperature chamber) to the circuit card (inside the chamber). Cable length should be less than two (2) feet, if possible.

(b) Turn off both power supplies.
(c) Install A9 card in test set.
(d) Turn on both the 20 Vdc and 5 Vdc power supplies, and adjust to 19.9 $\pm$0.1 Vdc and 4.9 $\pm$0.1 Vdc, respectively, using the DVMs.
(e) Adjust the potentiometer on the A9 card to the minimum exit delay setting (full clockwise).
(f) Place S1, S4, S5, and S6 in Position 1, and momentarily depress reset switch S3.

PASS - All test indicators except the ACC LED are OFF, and the ACC LED is turned ON.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

FAIL - Any test set indicator except the ACC LED turned ON or the ACC LED is turned OFF.

NOTE: If the yellow test-in-progress indicator is ON, wait until it turns OFF before depressing S3.

(g) Wait to allow the A9 card circuitry to stabilize - approximately 30 seconds.
(h) Verify that the yellow test-in-progress indicator is OFF.
(i) Momentarily depress S2, and observe test set low and high limit indicators.

PASS - Yellow test-in-progress indicator will light, and before it extinguishes both the low and high limit green indicators will light and will remain lit when the test in progress indicator turns off.

FAIL - Either or both low or high limit red indicators light.

(j) Wait until the test-in-progress indicator goes OFF; place S5 in Position 2; and momentarily depress reset switch S3.
(k) Repeat Steps (h) and (i).
(l) Wait until the test-in-progress indicator goes OFF; place S5 in Position 3; and momentarily depress reset switch S3.
(m) Repeat Steps (h) and (i).
(n) Wait until the test-in-progress indicator goes OFF; then depress S7, and observe the IA1 LED.

PASS - The LED is OFF before depressing S7; the LED turns ON while S7 is depressed; and the LED turns OFF when S7 is released.

FAIL - The LED is ON before depressing S7; the LED remains OFF while the switch is depressed; or the LED remains ON when the switch is released.

(o) Momentarily depress reset switch S3.
(p) Verify that the yellow test-in-progress indicator is OFF; momentarily depress S2; then immediately hold S8 depressed and observe the test-in-progress, the green high limit and IA2 LED's.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:10Z
Check the source to verify that this is the currentversion before use.

PASS - The yellow test-in-progress indicator will turn ON; before the green high limit indicator lights, the IA2 LED will be ON while S8 is depressed; and when the green high limit indicator lights, the IA2 LED will turn OFF while S8 is depressed.

FAIL - The IA2 LED remains OFF before the green high limit indicator lights or the IA2 LED remains ON when the green high limit indicator lights.

(q) Wait until the test-in-progress indicator goes OFF; depress S9, and observe the NL LED.

PASS - The LED is OFF before S9 is depressed; the LED is ON when S9 is depressed; and the LED is OFF when S9 is released.

FAIL - The LED is ON before S9 is depressed; the LED remains OFF when S9 is depressed; or the LED is ON when S9 is released.

(r) Place S4 in Position 2 and momentarily depress reset switch S3.

PASS - See Step (f).

FAIL - See Step (f).

(s) Place S6 in Position 2, and observe the test-in-progress, low and high limit and SEC LED's.

PASS - Both the yellow test-in-progress and green SEC LED's will light when S6 is placed in Position 2. Before the test-in-progress indicator goes OFF, both low and high limit green indicators will turn ON.

FAIL - Either or both low or high limit indicators turn ON.

(t) Wait until the test-in-progress indicator goes OFF; place S6 in Position 1 and S1 in Position 2.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:10Z
Check the source to verify that this is the current version before use.

(u) Verify that the test-in-progress indicator is OFF, and adjust the potentiometer on the A9 card to the maximum exit delay setting (full counterclockwise).

(v) Momentarily depress reset switch S3.

PASS - See Step (f).

FAIL - See Step (f).

(w) Repeat Step (s).

NOTE: This test step will take approximately 2-1/2 minutes. Wait until the yellow test-in-progress indicator goes OFF before proceeding to Step (x).

(x) Depress S10 and observe the EI LED.

PASS - The LED is OFF before depressing S10; the LED is ON when S10 is depressed; and the LED is OFF when S10 is released.

FAIL - The LED is ON before S10 is depressed; the LED remains OFF when S10 is depressed; or the LED remains ON when S10 is released.

(y) Turn off both power supplies.
(z) Remove the A9 card from the test set.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

TEST METHOD 11

LATCHED ALARM PRINTED WIRING BOARD ASSEMBLY
(13220E4217)

A10 CARD

FUNCTIONAL TEST

1. TEST APPARATUS:

(a) Power supply, Hewlett-Packard 6202B or equivalent (two required).
(b) Digital voltmeter, Fairchild 7000A or equivalent (two required).
(c) Test fixture (see Figure A-15).
(d) Test fixture schematic A (see Figure A-16).
(e) Test fixture schematic B (see Figure A-17).
(f) A10 card test set-up (see Figure A-18).

2. TEST PROCEDURE:

(a) Connect test apparatus as shown in Figure A-18.

NOTE: Test fixture shall be kept at room ambient temperature, with no more than a $\pm 5^{\circ}$ F swing during tests, without recalibrating the fixture. When performing functional test as a part of the low or high temperature test, a connecting cable must be run from the test fixture (outside the temperature chamber) to the circuit card (inside the chamber). Cable length shall be less than two (2) feet, if possible.

(b) Turn off both power supplies.
(c) Install A10 card in test set, and plug in test point adapter.
(d) Turn on both the 20 Vdc and 5 Vdc power supplies, and adjust to 19.9 $\pm$0.1 Vdc and 4.9 $\pm$0.1 Vdc, respectively, using the DVM's.
(e) Adjust the potentiometer on the A10 card to the minimum entrance delay setting (full clockwise).
(f) Place both S1 and S4 in Position 1, and momentarily depress both reset switches S3 and S6.

PASS - All test set indicators are OFF.

FAIL - Any test set indicator turned ON.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

NOTE: If the yellow test-in-progress indicator is ON, wait until it turns OFF before depressing S3 and S6.

(g) Wait to allow the A10 card circuitry to stabilize - approximately 30 seconds.
(h) Verify that the yellow test-in-progress indicator is OFF.
(i) Momentarily depress S2, and observe test set low and high limit indicators.

PASS - Yellow test-in-progress indicator will light, and before it extinguishes both the low and high limit green indicators will light and will remain lit when the test in progress indicator turns off.

FAIL - Either or both low and high limit red indicators light.

(j) Wait until the test-in-progress indicator goes off, adjust the potentiometer on the A10 card to the maximum entrance delay setting (full counterclockwise).
(k) Place S1 in Position 2, and momentarily depress S3.

PASS - All low limit and high limit indicators turn OFF.

FAIL - Any low limit or high limit indicator remains ON.

NOTE: If the yellow test-in-progress indicator is ON, wait until it extinguishes before depressing S3.

(l) Verify that the yellow test-in-progress indicator is out.
(m) Momentarily depress S2, and observe test set low and high limit indicators.

PASS - See Step (i).

FAIL - See Step (i).

NOTE: This test step will take approximately 2-1/2 minutes. Wait until the yellow test-in-progress indicator goes OFF before proceeding to Step (n).

(n) Momentarily depress S6.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

PASS - Both the LAA and DIT LED's turn OFF.

FAIL - Either or both the LAA and DIT LED's remain ON.

(o) Momentarily depress S5.

PASS - Both the LAA and DIT LED's turn ON and remain ON when S5 is released.

FAIL - Either or both the LAA and DIT LED's fail to turn ON and remain ON.

(p) Place S4 in Position 2.
(q) Repeat Step (n).
(r) Momentarily depress S5.

PASS - Only the DIT LED turns ON and remains ON when S5 is released.

FAIL - The DIT LED remains OFF or the LAA LED lights.

(s) Momentarily depress S7 and observe MR1 LED.

PASS - The LED is OFF before depressing the switch; the LED turns ON while the switch is depressed; and the LED turns OFF when the switch is released.

FAIL - The LED is ON before depressing the switch; the LED remains OFF while the switch is depressed; or the LED remains ON when the switch is released.

(t) Momentarily depress S8 and observe the MR2 LED.

PASS - See Step (s).

FAIL - See Step (s).

(u) Turn off both power supplies.
(v) Remove the test point adapter, and remove the A10 card from the test set.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

TEST METHOD 12

ALARM I.D. PRINTED WIRING BOARD ASSEMBLY
(13220E4218)

A11 CARD

AND

LED PRINTED WIRING BOARD ASSEMBLY
(13220E4219)

A12 CARD

FUNCTIONAL TEST

1. TEST APPARATUS:

   (a) Power supply, Hewlett-Packard 6202B or equivalent (two required).
   (b) Digital voltmeter, Fairchild 7000A or equivalent (two required).
   (c) Test fixture (see Figure A-19).
   (d) Test fixture schematic (see Figure A-20).
   (e) A11/A12 card test set-up (see Figure A-21).

2. TEST PROCEDURE A12/A11 CARDS:

   (a) Connect test apparatus as shown in Figure A-21.

   NOTE: Test fixture shall be kept at room ambient temperature, with no more than a ±5° F swing during tests, without recalibrating the fixture. When performing functional test as a part of the low or high temperature test, a connecting cable must be run from the test fixture (outside the temperature chamber) to the circuit card (inside the chamber). Cable length shall be less than two (2) feet, if possible.

   (b) Turn off both power supplies.
   (c) Install A12 card to be tested in test fixture.
   (d) Turn on the 20 Vdc supply; adjust voltage to 19.9 ±0.1 Vdc using the DVM; depress S5 and observe LED's.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

PASS - All LED's lit with the same apparent brightness with no obvious angular misalinement.

FAIL - Any LED unlit; visually apparent differences between lighted LED's; or obvious misalinement of any LED's.

(e) Turn off 20 Vdc power supply.
(f) Remove A12 card from test fixture, and install A12 card onto the A11 card to be tested.
(g) Install A11 card to be tested in test fixture.
(h) Turn on both the 20 Vdc and 5 Vdc power supplies, and adjust to 19.9 $\pm$0.1 Vdc and 4.9 $\pm$0.1 Vdc, respectively, using the DVM's.
(i) Place S1 in Test 1 position.
(j) Momentarily depress reset switch S3 and observe amber test fixture light and A12 card lights.

PASS - Amber test fixture light will turn ON for approximately 5 seconds, and A12 LED's will either turn OFF or remain OFF.

FAIL - A12 LED's remain ON or intermittent after the amber test fixture light turns OFF.

(k) Momentarily depress S2, and observe A12 LED's.

PASS - LED's F, A, B, C, D, and E will light in sequence in a period of approximately 1 second, and will remain lit with uniform apparent brightness.

FAIL - Any LED intermittent or unlit after period of approximately 1 second or obvious visual differences in brightness between the LED's.

(l) Place S1 in Test 2 position and S4 in "F" position.
(m) Momentarily depress reset switch S3 and observe amber test fixture light and A12 card LED's.

PASS - See Step (g).

FAIL - See Step (j).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

(n) Momentarily depress S2 and observe green test fixture light and A12 card LED's.

PASS - Only the "F" LED will light and remain lit; the green test fixture light will be ON at the end of the test.

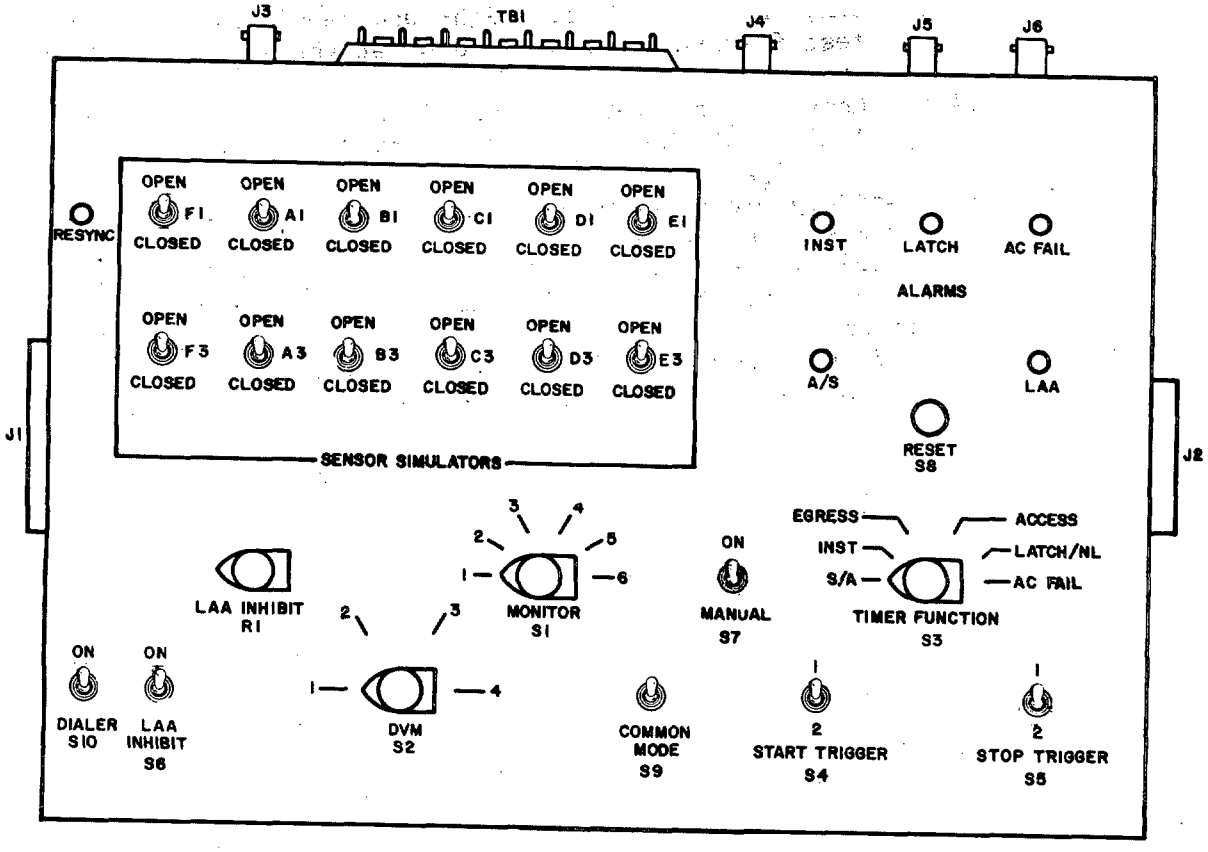
FAIL - Momentary blinking or turn ON of other LED's or failure of the "F" LED to light.

(o) Successively place S4 in Positions A, B, C, D, and E, and repeat Steps (m) and (n). Note, only the single LED indicated by the setting of S4 should light. Activation or momentary blinking of other LED's shall be cause for rejection.

(p) Turn off both power supplies.

(q) Remove the A11 card from the test fixture.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.



NOTE:
SEE FIGURE A-2 FOR TEST FIXTURE SCHEMATIC DIAGRAM.

FIGURE A-1. Control unit test fixture.

X-2978B

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

X-2979C

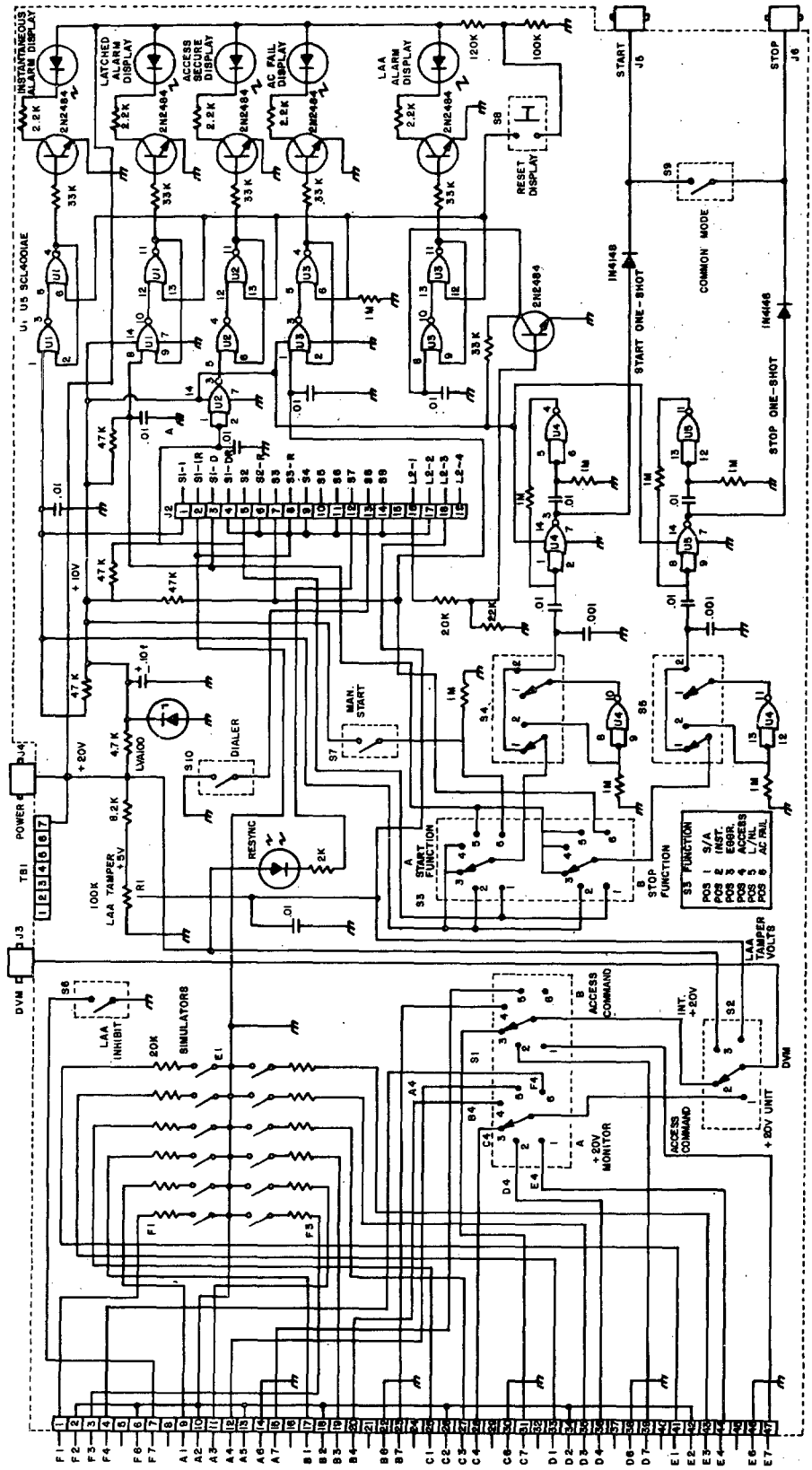


FIGURE A-2. Control unit test fixture schematic.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

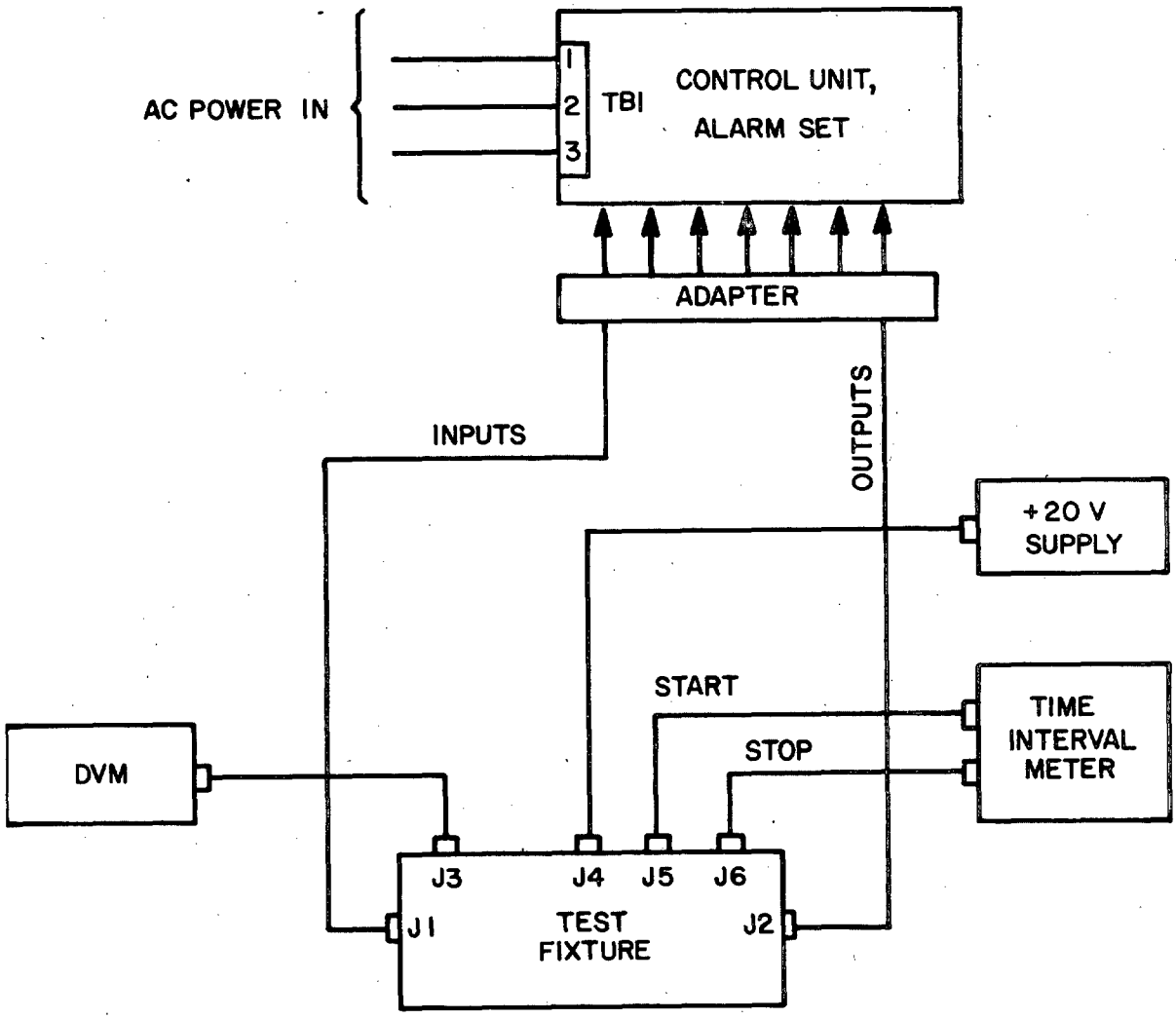


FIGURE A-3. Control unit test setup.

X-2980A

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

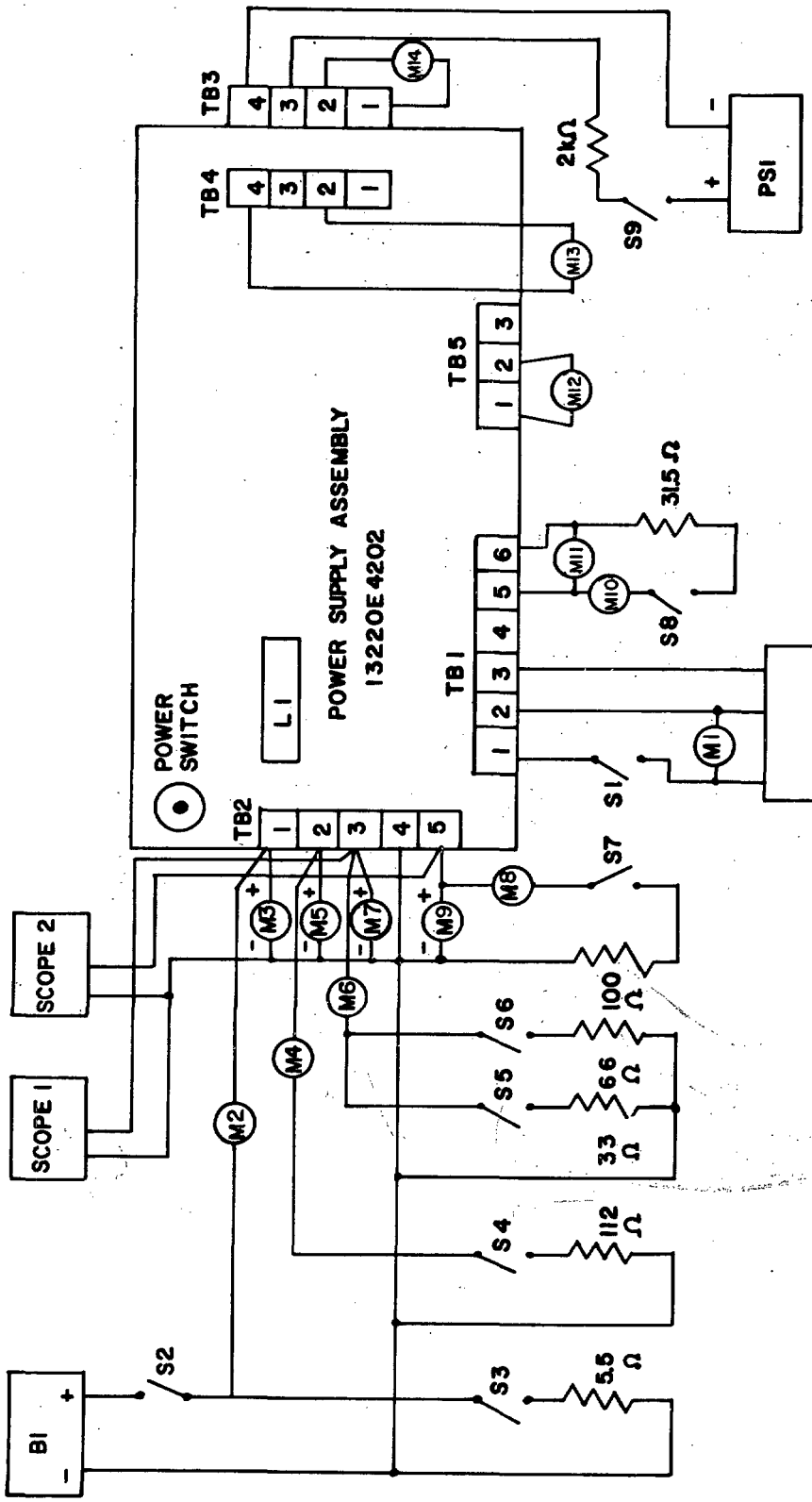


FIGURE A-4. Power supply assembly test fixture.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

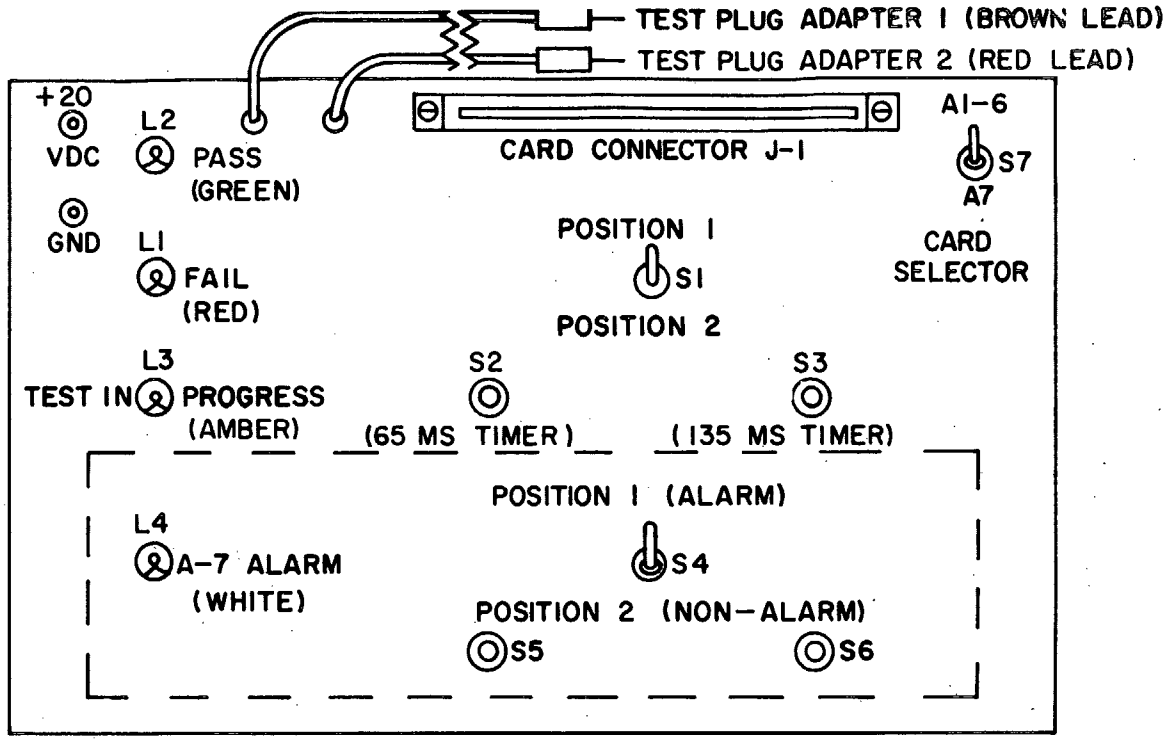
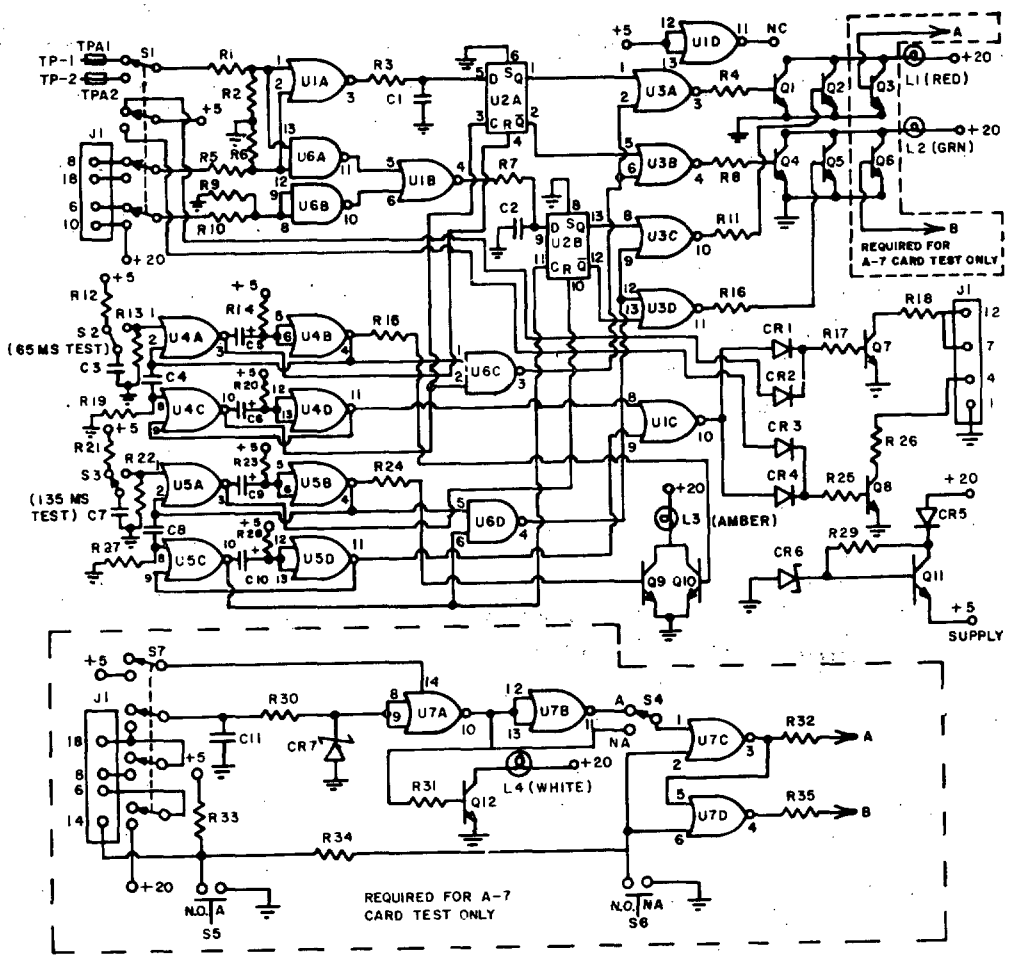


FIGURE A-5. A1-A6 and A7 cards test fixture.

X-3911

MIL-C-52853B
APPENDIX

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.



U#	TYPE
1,3,4,5,7	#4001 QUAD 2 INPUT NOR GATE
2	#4013 DUAL "D" FLIP-FLOP W/SET/RESET
6	#4011 QUAD 2 INPUT NAND GATE .

ALL CAPACITORS LOW LEAKAGE MYLAR, POLYCARBONATE, OR TANTALUM (±10% TOL.)

C#	VALUE	VOLTAGE
1,2	2 uF	12V
3,4,7,8	.01 uF	12V
5,9	43 uF	12V
6,10	2.2 uF	12V
11	.001 uF	25V

S#	TYPE
1,7	4PDT TOGGLE, CUTLER-HAMMER #SF4SCY191 OR EQ.
2,3	MOM. SPDT (BBM), GRAYHILL #46-101 OR EQ.
4	SPDT TOGGLE, CUTLER-HAMMER #8282K14 OR EQ.
5,6	MOM. SPST-N.O., GRAYHILL #39-1 OR EQ.

CR#	TYPE
1,2,3,4,5	IN456 OR EQ.
6	IN753 - 6.2V ZENER
7	IN751 - 5.1V ZENER

ALL RESISTORS 1/4W, ±10% TOL. (UNLESS NOTED).

R#	VALUE
1,5,10,30	15K
2,6,9	4.7K
3	NOTE 1
4,8,11,15,16,17,24,25, 31,32,35	10K
7	NOTE 2
12,13,19,21,22,27,34	47K
14,23	NOTE 3
18,26	20K
20	NOTE 4
28	NOTE 5
29	1K (1/2 WATT)
33	680Ω

J1 - CARD CONNECTOR, 18 PIN, ELCO 6007 OR EQ.

Q1 THRU Q11 - 2N2222A OR EQ.

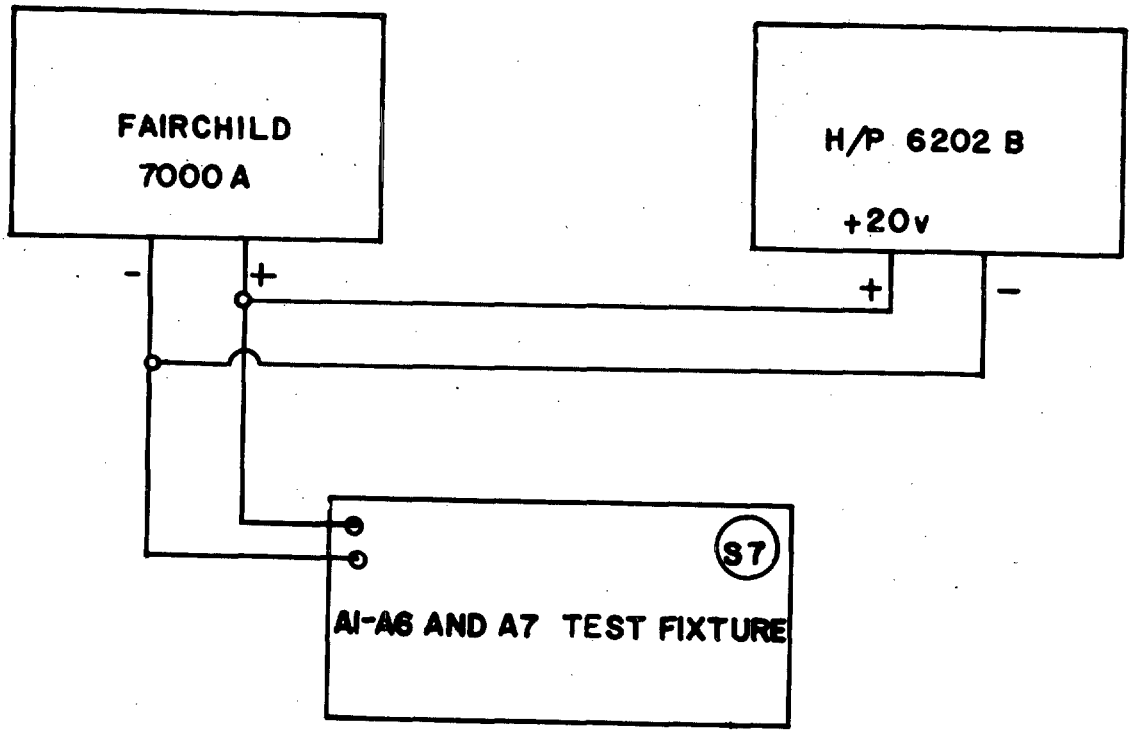
L1 THRU L4 - #327, SYLV. #33309-0 OR EQ.

TPA 1, 2 - TIP PLUG, SMITH #492 OR EQ.

NOTE 1: R3 ≅ 13K, TBD SUCH THAT A 10.00 ± 0.05 MS AND GREATER OUTPUT AT U1A CAUSES U2A Q = 0 VDC AND U2A $\overline{Q}$ = +5VDC, WHEN U2A-C IS STROBED AT 100 KHz OR GREATER.
NOTE 2: R7 ≅ 21.5K, TBD SUCH THAT A 20.00 ± 0.10 MS AND GREATER OUTPUT AT U1B CAUSES U2B Q = +5VDC AND U2B $\overline{Q}$ = 0 VDC, WHEN U2B-C IS STROBED AT 100 KHz OR GREATER.
NOTE 3: R14, R23 ≅ 100K, SELECT FOR TIMER OUTPUT ≅ 2 SECONDS.
NOTE 4: R20 ≅ 70K, SELECT FOR 65± 2 MS PULSE DURATION AT U4D OUTPUT.
NOTE 5: R28 ≅ 94K, SELECT FOR 135±4 MS PULSE DURATION AT U5D OUTPUT.
NOTE 6: CIRCUIT TRACE LENGTH SHOULD BE MINIMUM PRACTICABLE TO AVOID CIRCUIT OSCILLATION.

FIGURE A-6. A1-A6 and A7 cards-test fixture schematic. X-3912

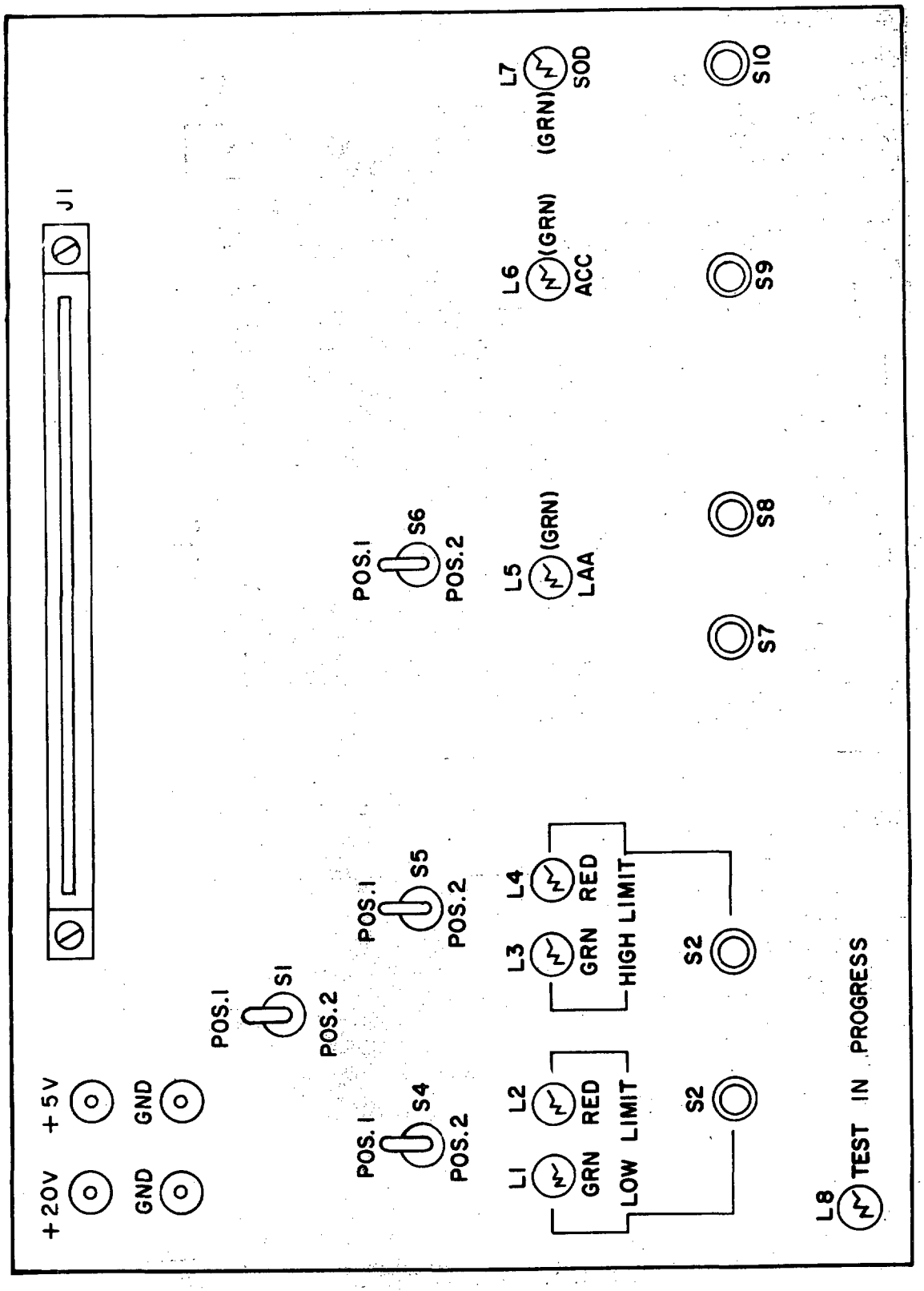
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.



NOTE: SWITCH S7 ON TEST SET MUST BE IN THE A1-A6 POSITION FOR TEST OF A1-A6 CARDS, AND IN THE A7 POSITION FOR A7 CARD TEST.

FIGURE A-7. A1-A6 and A7 card test set-up.

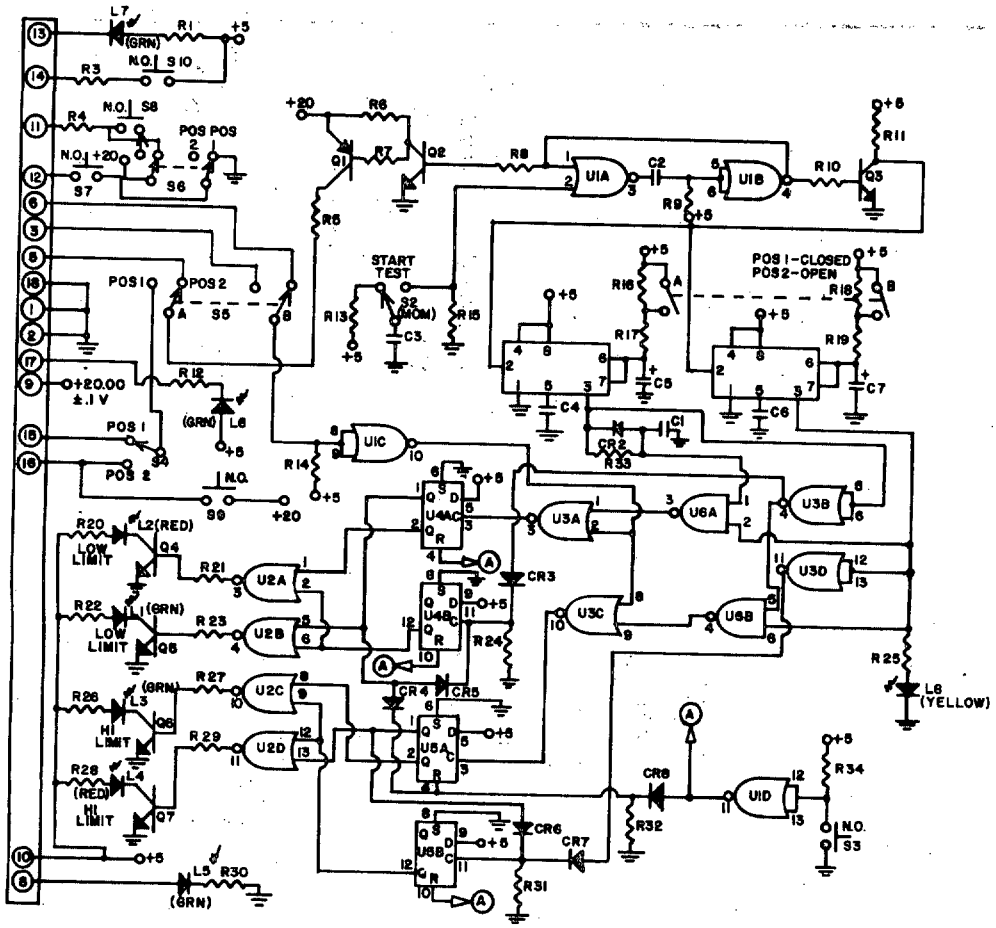
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.



X-3914

FIGURE A-8. A8 card test fixture.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.



R#	VALUE
1,12,20,22,25,26,28	470Ω
3,4,5,30	1K
6,7,11,14,21,23,27,29	10K
8,10,13,15,24,31,32,33,34	47K
9	NOTE 1
16,17	NOTE 2
18,19	NOTE 3

ALL RESISTORS 1/4W, ±10% TOLERANCE UNLESS NOTED.

S#	TYPE
1	DPST TOGGLE, ALCO SWITCH #MCT-210N OR EQ.
2	SPDT MOM. GRAYHILL #46-101 OR EQ.
4	SPDT TOGGLE
3,7,8,9,10	SPST MOM. N.O., GRAYHILL #39-1 OR EQ.
5,6	DPDT TOGGLE, CUTLER-HAMMER #SF2SCY191 OR EQ.

CR#	TYPE
1	IN751 ZENER OR EQ.
2-8	IN456 OR EQ.

Q#	TYPE
[illegible]	2N2907A OR EQ.
4-7	2N2222A OR EQ.

U#	TYPE
1,2,3	#4001 QUAD 2-INPUT NOR GATE
4,5	#4013 DUAL "D" FLIP-FLOP W/SET/RESET
6	#4011 QUAD 2-INPUT NAND GATE
7,8	#555 TIMER

NOTE: TIE ALL UNUSED GATE INPUTS TO +5 VOLTS.

C#	VALUE
1	2.2μF
2	0.1μF
3,4,6	0.01μF
5,7	45μF

ALL CAPACITORS ARE 12 WVDC LOW LEAKAGE MYLOR, POLYCARBONATE OR TANTALUM (±10% TOLERANCE).

L#	TYPE
1,3,5,6,7	MV5253 (GREEN) OR EQ.
2,4	MV5053 (RED) OR EQ.
8	MV5353 (YELLOW) OR EQ.

J1- CARD CONNECTOR, 18 PIN, ELCO #6007 OR EQ.

NOTES:
1. R9 ≅ 390K, SELECTED FOR U1B OUTPUT OF 20.0 ± 1.0 MS.
2. R16 ≅ 75K, SELECTED FOR U7 OUTPUT OF 8.0 ± 0.1 SECONDS, R17 ≅ 20K, SELECTED FOR U7 OUTPUT OF 4.0 + 0.1, -0.0 SECONDS.
3. R18 ≅ 100K SELECTED FOR U8 OUTPUT OF 12.0 ± 0.1 SECONDS; R19 ≅ 125K, SELECTED FOR U8 OUTPUT OF 7.0 + 0.1, -0.0 SECONDS.
4. CIRCUIT TRACE LENGTH SHOULD BE THE MINIMUM PRACTICABLE TO AVOID CIRCUIT OSCILLATION.

FIGURE A-9. A8 card-test fixture schematic. X-3915

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

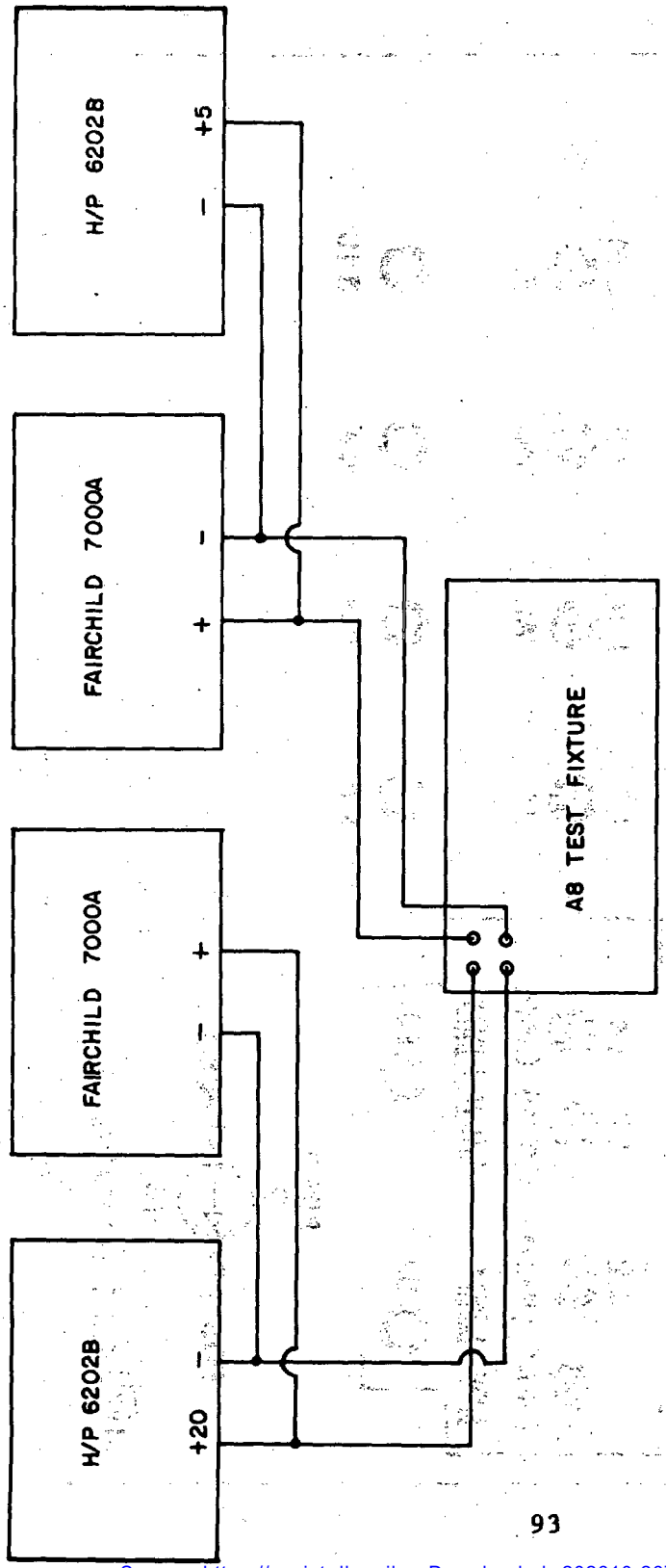


FIGURE A-10. A8 card test set-up.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

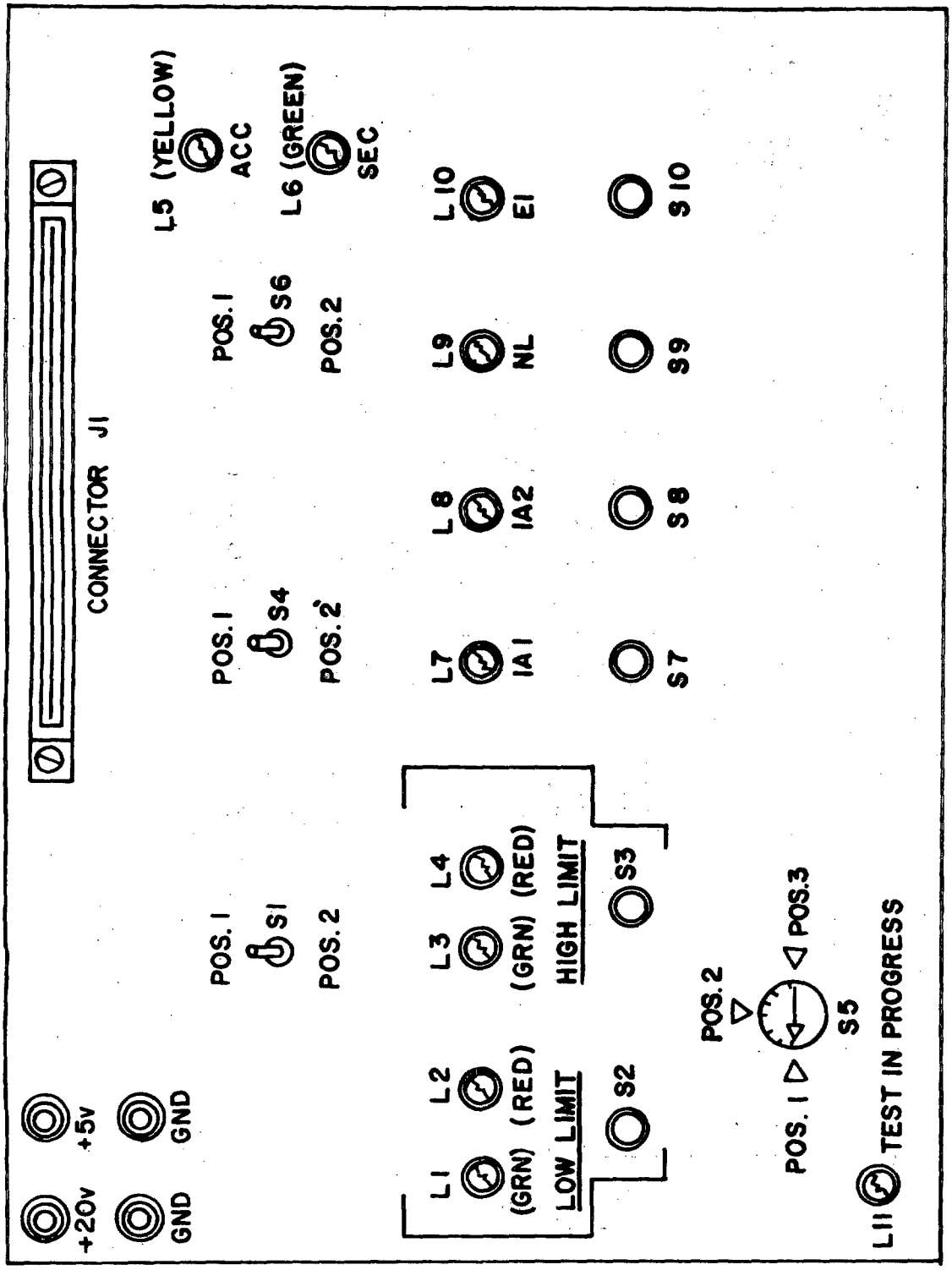
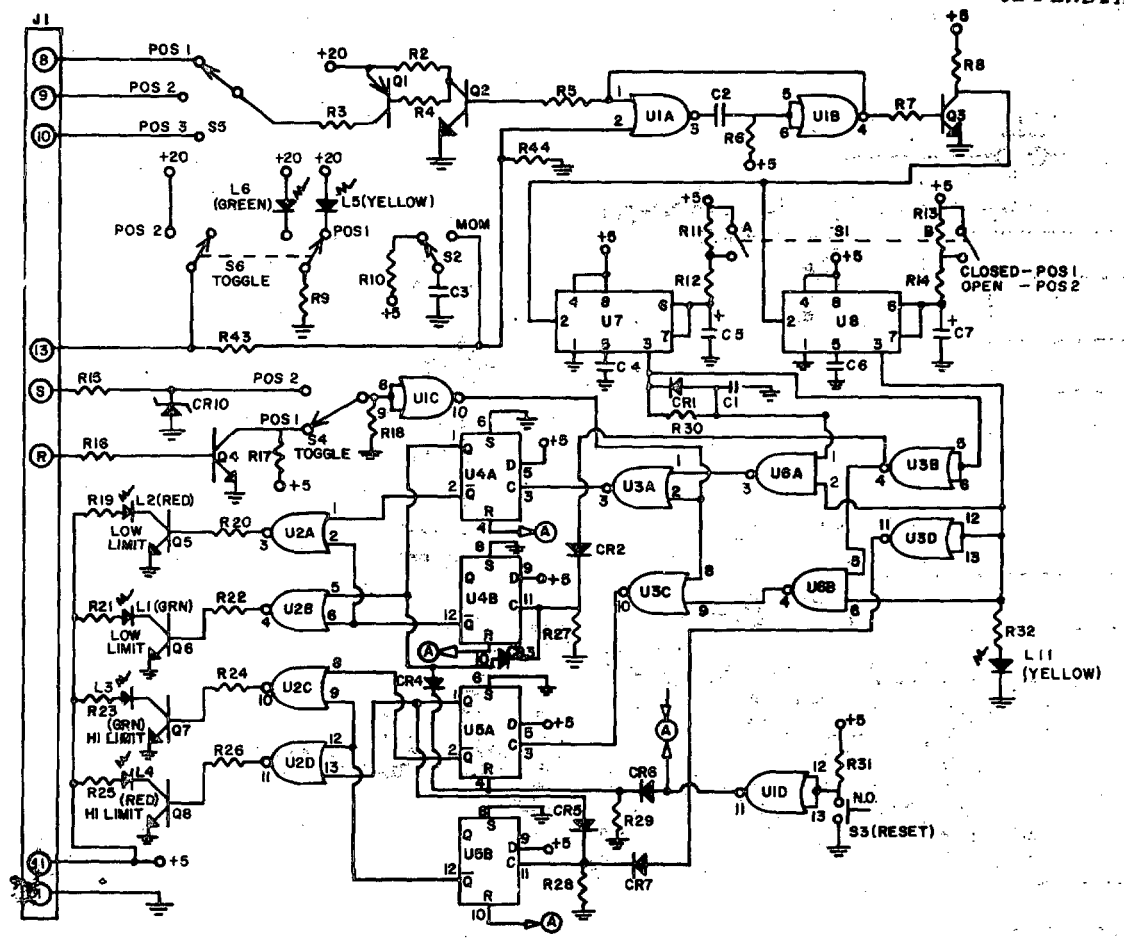


FIGURE A-11. A9 card test figure

X-3917

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.



U#	TYPE
1,2,3,9	#4001 QUAD 2-INPUT NOR GATE
4,5	#4013 DUAL "D" FLIP-FLOP W/SET/RESET
6	#4011 QUAD 2-INPUT NAND GATE
7,8	#555 TIMER

NOTE: TIE ALL UNUSED GATE INPUTS TO +5 VOLTS

L#	TYPE
1,3,6	MV5253 (GREEN) OR EQ.
2,4,7,8,9,10	MV5053 (RED) OR EQ.
5,11	MV353 (YELLOW) OR EQ.

S#	TYPE
1	DPST TOGGLE, ALCOSWITCH #MCT-210N OR EQ.
2,4	SPDT MOM. GRAYHILL #46-101 OR EQ.
3,7,8,9,10	SPST MOM. N.O., GRAYHILL #39-1 OR EQ.
5	3 POSITION ROTARY, GRAYHILL SERIES 50 OR EQ.
6	DPDT TOGGLE, CUTLER-HAMMER #SP2SCY191 OR EQ.

Q#	TYPE
1	2N2907 OR EQ.
2 THRU 12	2N2222A OR EQ.

CR#	TYPE
1 THRU 7	1N456 OR EQ.
10	1N751 ZENER OR EQ.

R#	VALUE
2,4,8,20,22,24,26,33,37,39	10K
3,9,17,41	1K
5,7,10,16,18,27,28,29,30,31,35,36,44	47K
6	NOTE 1
11,12	NOTE 2
13,14	NOTE 3
15	4.7K
19,21,23,25,32,34,38,40,42	470Ω
43	20K

ALL RESISTORS 1/4W, ±10% TOLERANCE UNLESS NOTED.

C#	VALUE
1	2.2μF
2	0.1μF
3,4,6	0.01μF
5,7	45μF

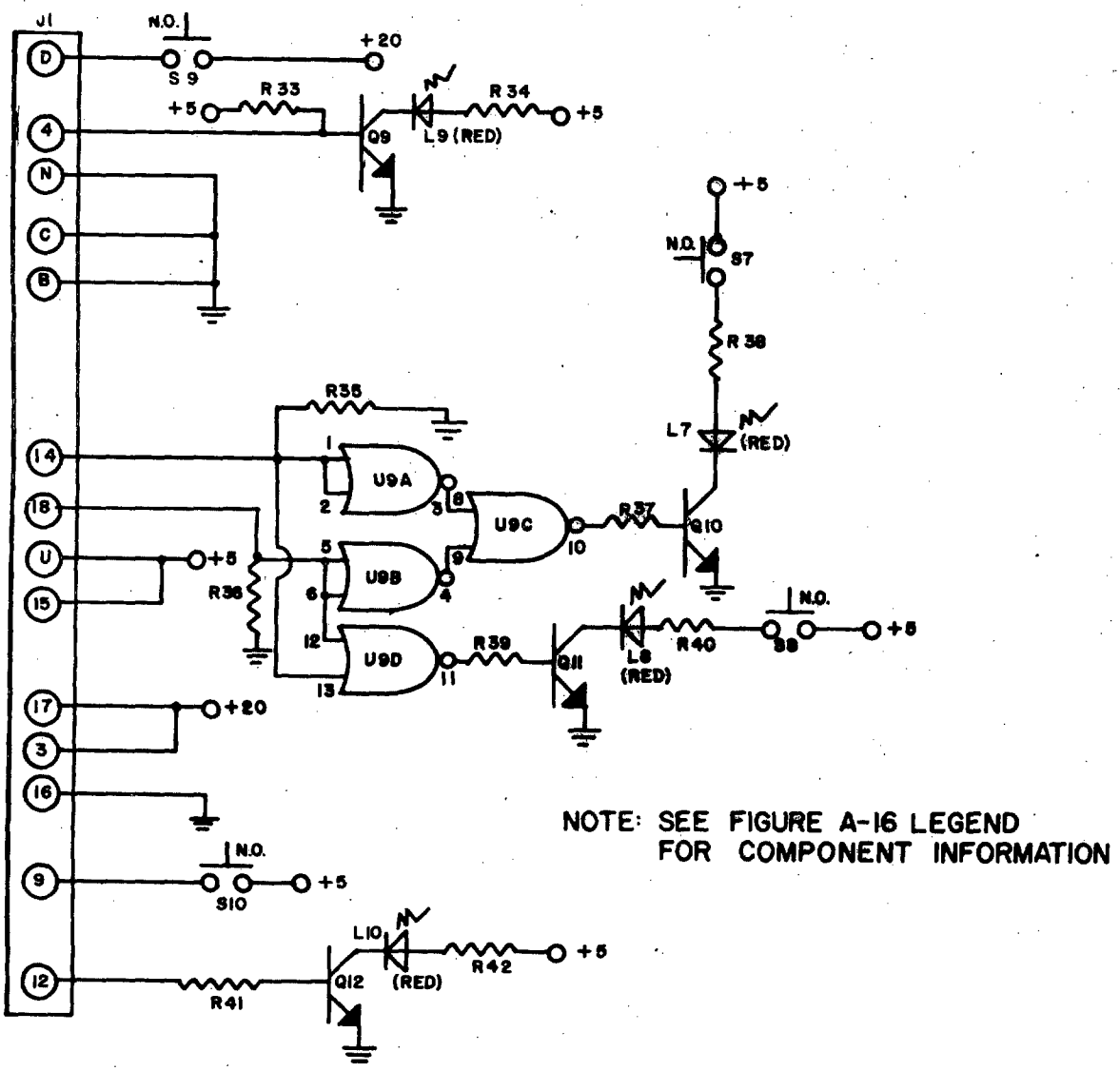
ALL CAPACITORS ARE 12 WVOC LOW LEAKAGE MYLOR, POLYCARBONATE OR TANTALUM (±10% TOLERANCE)

J1 CARD CONNECTOR, 36 PIN, AMPHENOL #225-27821-101 OR EQ.

NOTE: 1. R6 ≅ 1.5M, SELECTED FOR U1B OUTPUT OF 75.0 +1.0 MS.
2. R11 ≅ 1.4M, SELECTED FOR U7 OUTPUT OF 80±2 SECONDS. R12 ≅ 163K, SELECTED FOR U7 OUTPUT 8+0.0, -0.2 SECONDS.
3. R13 ≅ 2.35M, SELECTED FOR U8 OUTPUT OF 140±3 SECONDS. R14 ≅ 230K SELECTED FOR U8 OUTPUT 12.0 +0.2, -0.0 SECONDS.
4. CIRCUIT TRACE LENGTH SHOULD BE THE MINIMUM PRACTICABLE TO AVOID CIRCUIT OSCILLATION.

FIGURE A-12. A9 card test fixture schematic A. X-3918

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.



NOTE: SEE FIGURE A-16 LEGEND FOR COMPONENT INFORMATION

J-SIIDS A-9 CARD TEST FIXTURE
SCHEMATIC B – NON-TIMER FUNCTIONS

FIGURE A-13. A9 card test fixture schematic B. X-3919

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

X-3920

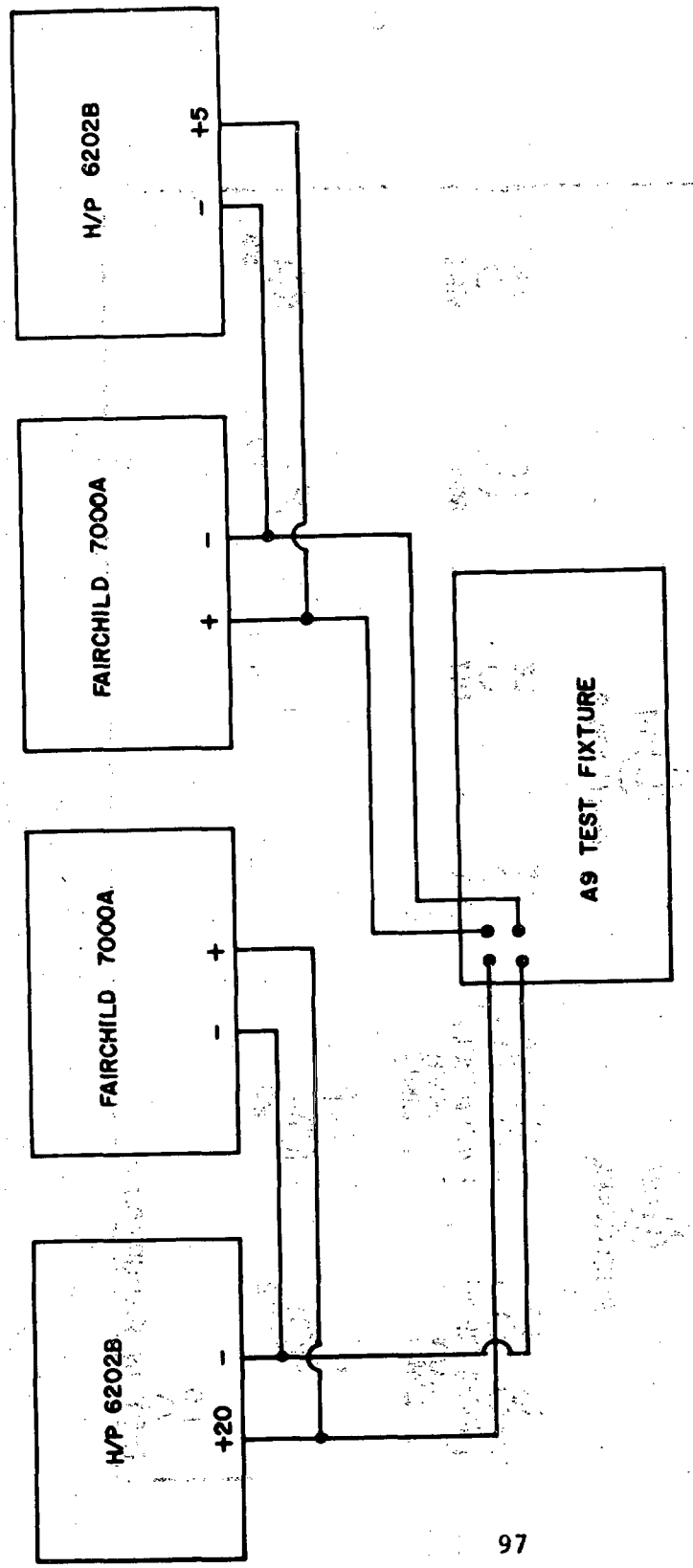
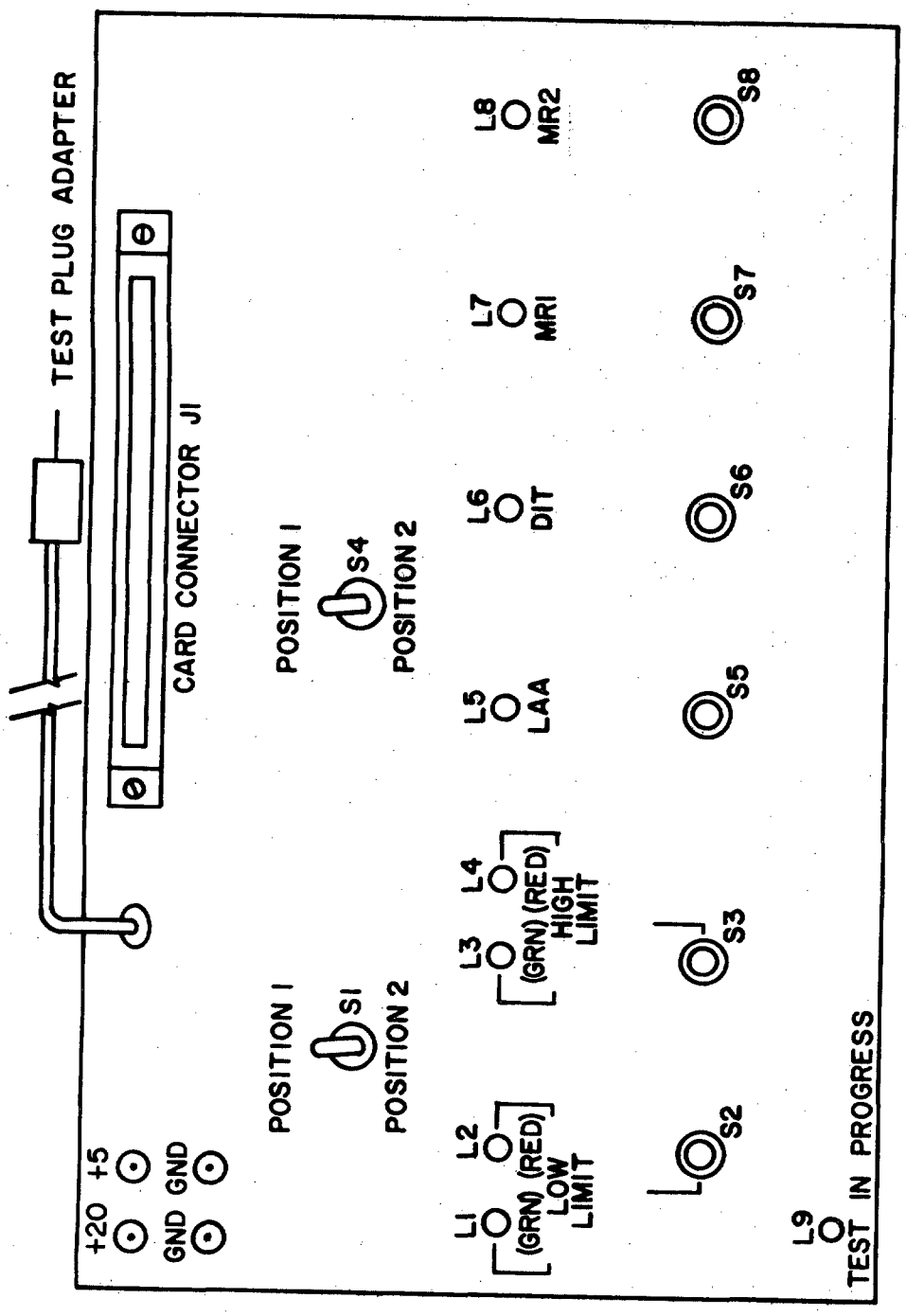


FIGURE A-14. A9 card test set-up.

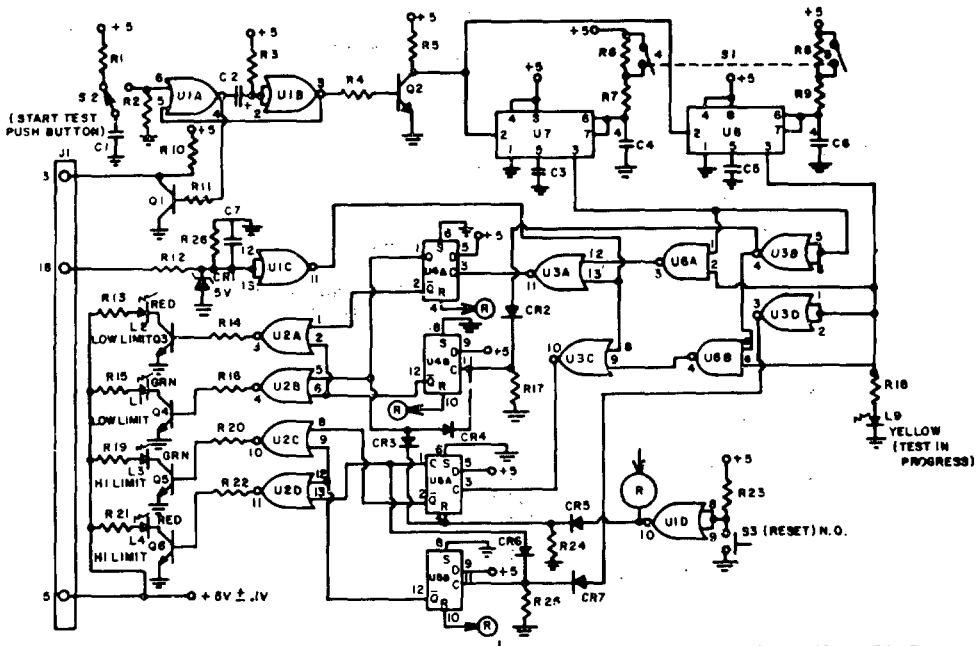
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.



X-3921

FIGURE A-15. AIO card test fixture.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.



U#	TYPE
1	#4001 QUAD 2-INPUT NOR GATE
2	#4001 QUAD 2-INPUT NOR GATE
3	#4001 QUAD 2-INPUT NOR GATE
4	#4013 DUAL "D" FLIP-FLOP W/SET/RESET
5	#4013 DUAL "D" FLIP-FLOP W/SET/RESET
6	#4011 QUAD 2-INPUT NAND GATE
7	#555 TIMER
8	#555 TIMER

ALL CAPACITORS LOW LEAKAGE MYLAR, POLY-CARBONATE OR TANTALUM (±10% TOL), 12 WVDC

C#	VALUE
1	.01μF
2	.1μF
3	.01μF
4	45μF TANTALUM
5	.01μF
6	45μF TANTALUM
7	4.5μF

S#	TYPE
1	TOG. DPST ALCO SWITCH #MCT-210N
2	MOM. SPDT, GRAYHILL #46-101 OR EQ.
3	MOM SPST N.O., GRAYHILL #39-1 OR EQ.

CR#	TYPE
1:	1N751-5.1V ZENER OR EQ.
2,3,4,5, 6,7:	1N456 OR EQ.

ALL RESISTORS 1/4W, ±10% TOLERANCE (UNLESS NOTED)

R#	VALUE
1,2,4,11,17,23, 24,25	47K
3	TBD NOTE 1
5,10,14,16,20,22	10K
6	TBD NOTE 2
7	TBD NOTE 2
8	TBD NOTE 3
9	TBD NOTE 3
12	4.7K
13,15,19,21	470Ω
18	330Ω
26	150K

L#	TYPE
1,3	MONSANTO MV 5253 OR EQ.
2,4,5,6, 7,8	MONSANTO MV 5053 OR EQ.
9	MONSANTO MV 5353 OR EQ.

J-1 CARD CONNECTOR AMPHENOL 225-27821-101 OR EQ.

Q1 THRU Q6 2N2222A OR EQ.

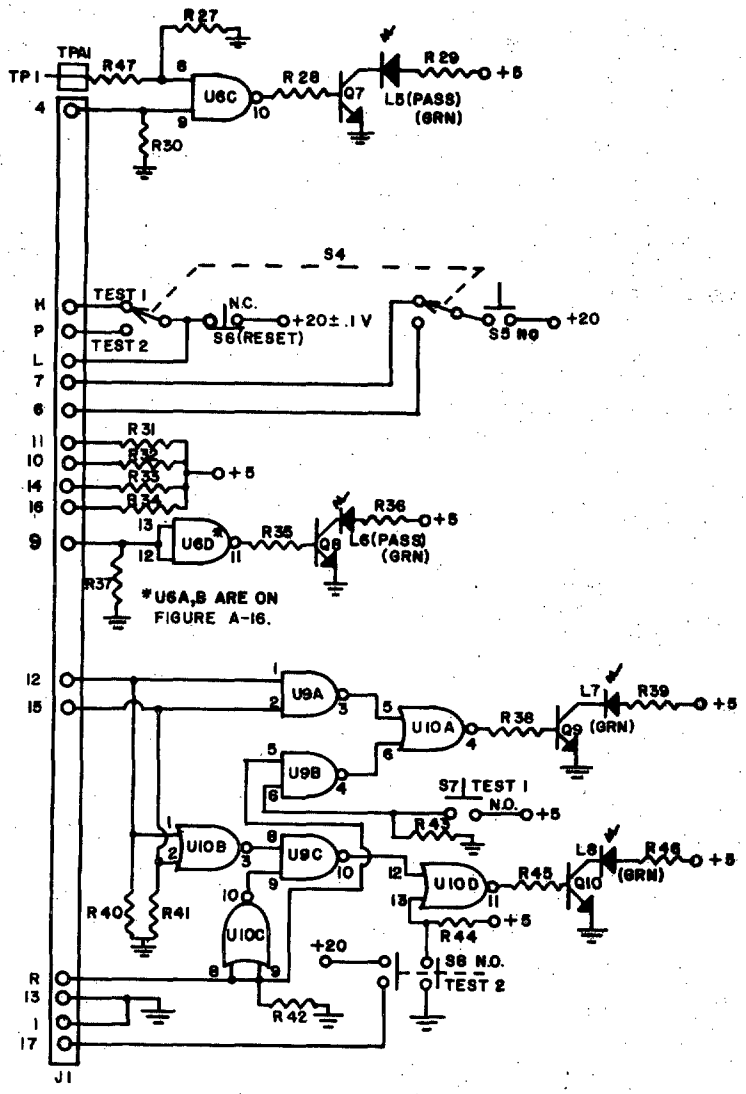
NOTE 1: R3 ≅ 390K. SELECTED FOR U1B OUTPUT AT 20.0±0.5 MS.
NOTE 2: R6 ≅ 1.4M. SELECTED FOR U7 OUTPUT AT 80 ±2 SECONDS. R7 ≅ 163K, SELECTED FOR U7 OUTPUT AT 8.0+0.0, -0.2 SECONDS.
NOTE 3: R8 ≅ 2.35M, SELECTED FOR U8 OUTPUT AT 140±3 SECONDS. R9 ≅ 230K, SELECTED FOR U8 OUTPUT AT 12.0 +0.2, -0.0 SECONDS.
NOTE 4: CIRCUIT TRACE LENGTH SHOULD BE THE MINIMUM PRACTICABLE TO AVOID CIRCUIT OSCILLATION.

NOTE 5: CIRCUITRY FOR TEST FIXTURE IS DEPICTED IN TWO SCHEMATICS, A AND B (FIGURE A-17), FOR CONVENIENCE ONLY. CONNECTOR J1 SHOWN ON THE THE TWO SCHEMATICS IS ONE AND THE SAME.

FIGURE A-16. A10 card test fixture schematic A.

X-3922A

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.



U#	TYPE
6,9	#4011 QUAD 2-INPUT NAND GATE
10	#4001 QUAD 2-INPUT NOR GATE

L#	TYPE
5,6,7,8	MONSANTO MV5253 (GREEN) OR EQ.

S#	TYPE
4	DPDT TOG. CUTLER-HAMMER #SF2SCY191 OR EQ.
5,7	MOM. SPST -N.O., GRAYHILL #39-1 OR EQ.
6	MOM. SPST -N.C., GRAYHILL #39-2 OR EQ.
8	MOM. DPST - N.O., GRAYHII #46-150 OR EQ.

NOTE 1: TIE ALL UNUSED GATE INPUTS TO +5 VOLTS.

ALL RESISTORS 1/4W, ±10% TOLERANCE (UNLESS NOTED)

R#	VALUE
27	4.7K
28,35,38,44,45	10K
29,36,39,46	330Ω
30,37,40,41,42,43	47K
31	270Ω
32,33,34	1K
47	15K

TPA 1 TIP PLUG, SMITH #492 OR EQ.

J1 - CARD CONNECTOR, 36 PIN, AMPHENOL #225-27821-101 OR EQ.

Q7 THRU 10 2N2222A OR EQ.

SCHEMATIC B- NON-TIMER FUNCTIONS

FIGURE A-17. AIO card test fixture schematic B.

X-3923A

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

X-3924

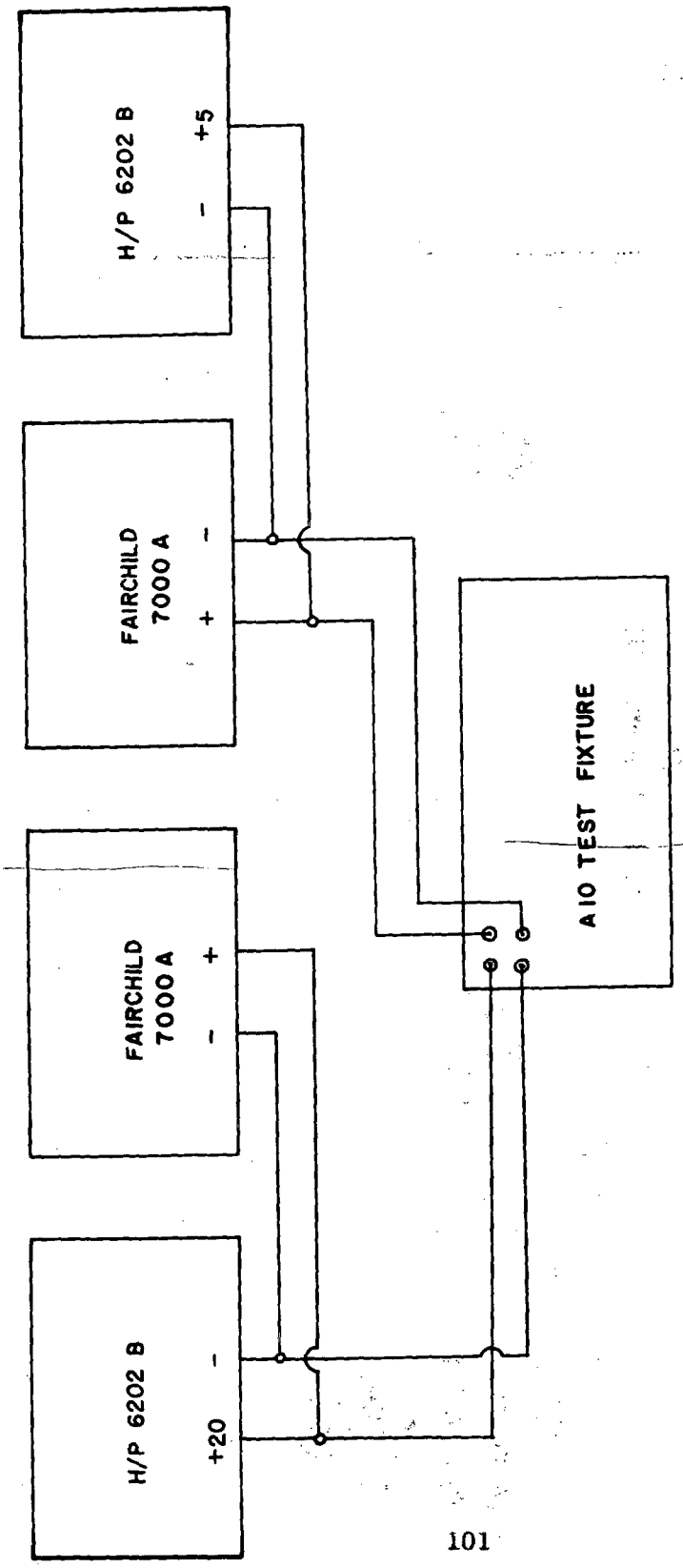


FIGURE A-18. A10 card test set-up.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

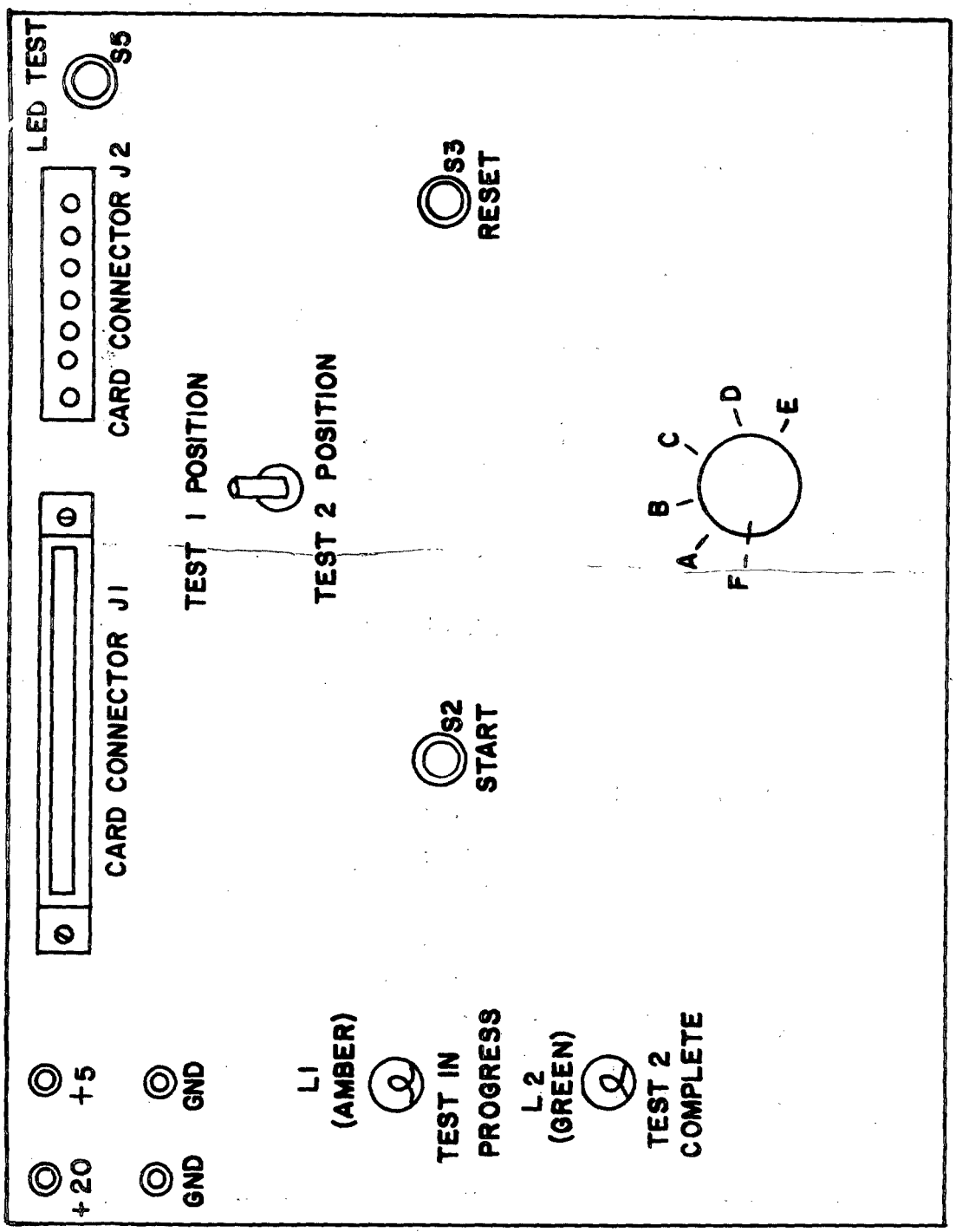
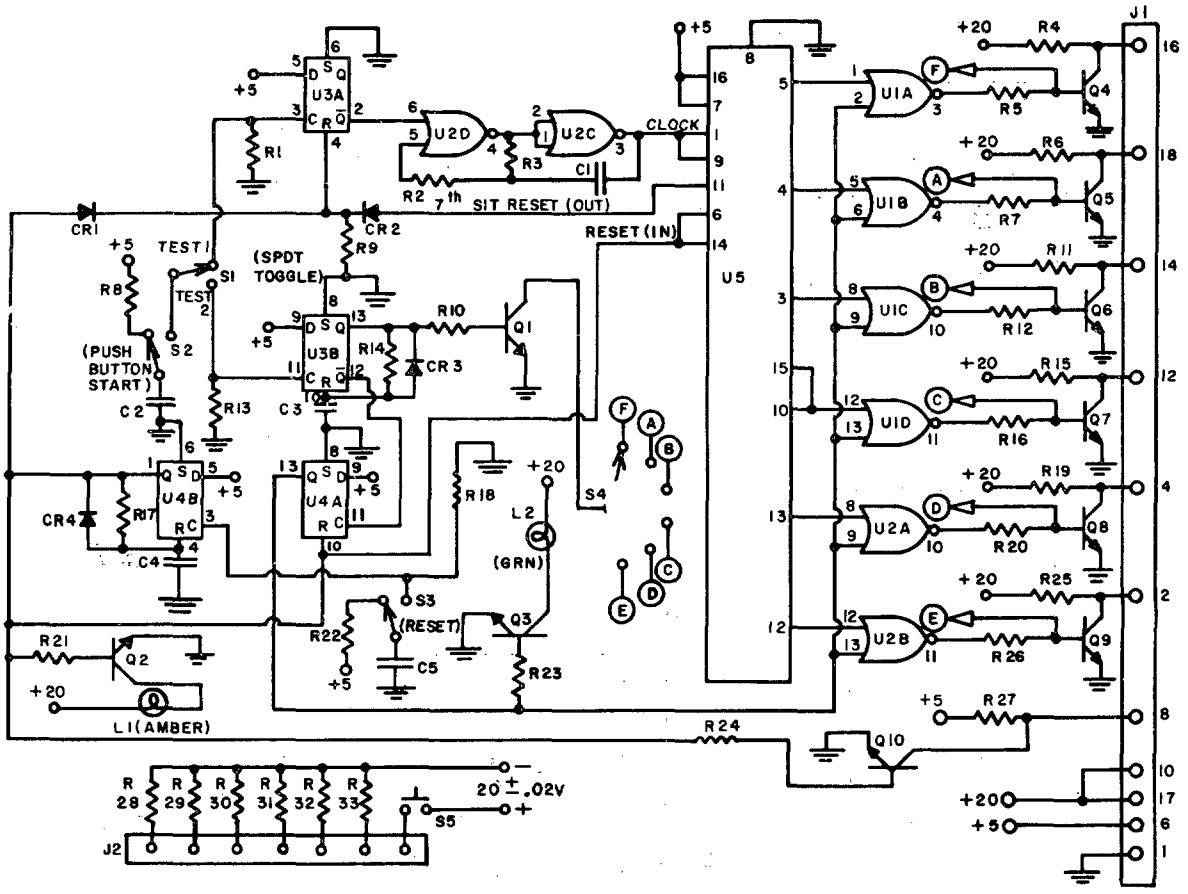


FIGURE A-19. A11/A12 cards test fixture.

X-3908

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.



U#	TYPE
1,2	#4001 QUAD 2 - INPUT NOR GATE
3,4	#4013 DUAL "D" FLIP-FLOP W/SET/RESET
5	#4015 STATIC SHIFT REGISTER

ALL CAPACITORS ARE LOW LEAKAGE MYLAR, POLYCARBONATE OR TANTALUM (±10% TOL.)

C#	VALUE	VOLTAGE
1	.1μF	12V
2,5	.01μF	12V
3	2μF	12V
4	10μF	12V

S#	TYPE
1	SPDT TOGGLE CUTLER-HAMMER #8282K14 OR EQ.
2	SPDT MOM. GRAYHILL #46-101 OR EQ.
3	MOM, SPDT GRAYHILL #46-101 OR EQ.
4	6 POSITION ROTARY, GRAYHILL SERIES 50 OR EQ.
5	N.O. SPST MOM. GRAYHILL #39-1 OR EQ.

ALL RESISTORS 1/4W, ±10% (UNLESS NOTED)

R#	VALUE
1,8,9,13,18,21,22,23,24	47K
2	NOTE 1
3	NOTE 2
4, 6, 11, 15, 19, 25	39K
5, 7, 10, 12, 16, 20, 26, 27	10K
14	NOTE 3
17	NOTE 4
28, 29, 30, 31, 32, 33	2.2K

J1 - CARD CONNECTOR, 18 PIN, ELCO 6007 OR EQ

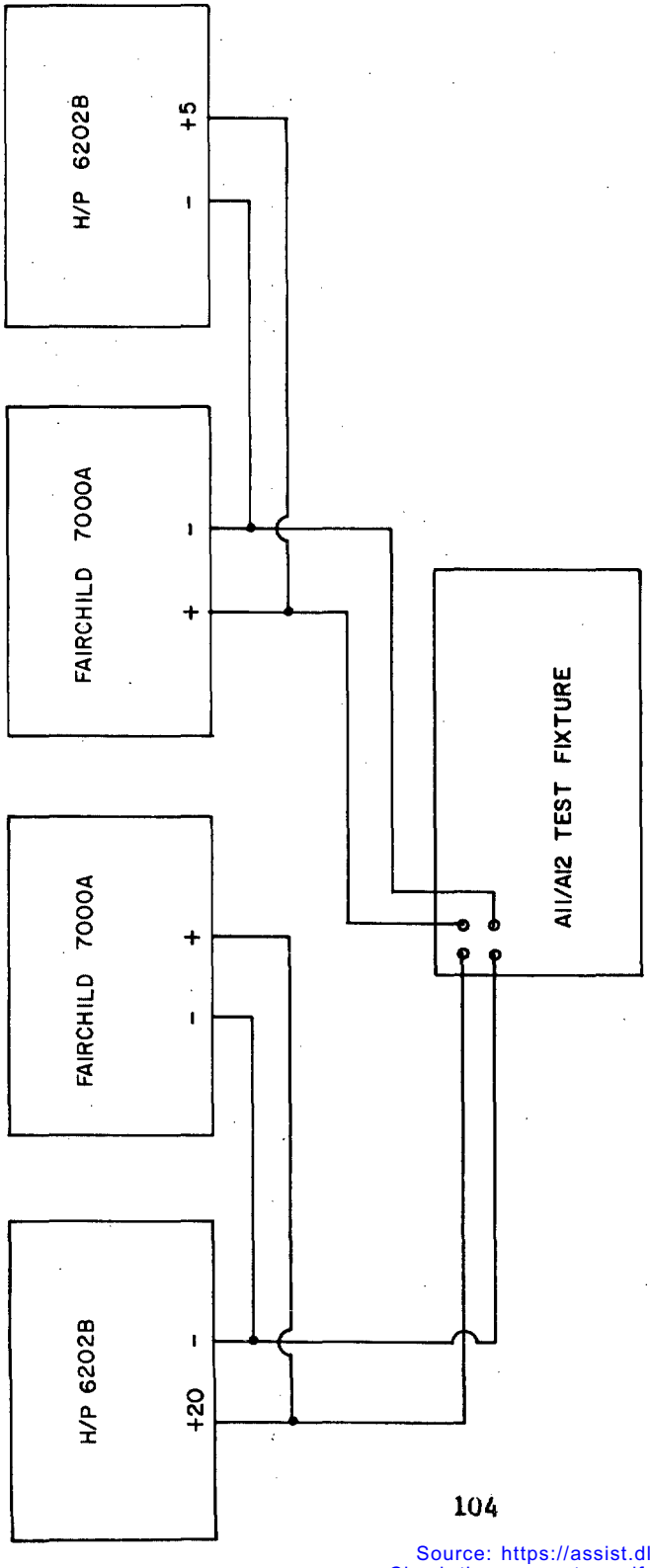
Q1 THRU Q10, 2N2222A OR EQ.

L1, L2, - #327SYLV, #33309-0 OR EQ.
CR1 THRU CR4 - IN456 OR EQ.

NOTE 1. APPROX. 4 TIMES R3
NOTE 2. TBD, APPROX. 560K, FOR U2C OUTPUT OF 6.25 Hz ±.2Hz.
NOTE 3. TBD, APPROX. 1M, FOR 1.5 SEC ±.1 SEC OUTPUT AT U3B-Q.
NOTE 4. TBD, APPROX. 820 K, FOR 5.0 SEC ± .5 SEC OUTPUT AT U4B-Q.
NOTE 5. ALL CIRCUIT TRACES SHOULD BE THE MINIMUM PRACTICAL TO AVOID CIRCUIT OSCILLATIONS

FIGURE A-20. A11/A12 cards test fixture schematic. X-3909

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:10Z
Check the source to verify that this is the current version before use.



X-3910

FIGURE A-21. A11/A12 cards test set-up.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

TABLE A-I. Control unit output conditions.

	INST	LATCH	S/A	AC FAIL	ACCESS COMMAND	LAA	+20V	COMMENTS
SECURE MODE								
INTRUSION	X[1]	X[2](X[4])	OFF	N/A	0 V	X[2]	20 ± 1	
TAMPER	X[1]	X[3](X[1])[OFF]	OFF	N/A	0 V	X[3][OFF]	20 ± 1	
DURESS	X[1]	X[3](X[1])	OFF	N/A	0 V	OFF	20 ± 1	
NORMAL (NON-ALARM	OFF	OFF	OFF	OFF	0 V	OFF	20 ± 1	
AC PWR "ON"	----	-----	---	OFF	----------	---	------	(AFTER 3 SEC DELAY, RESET BY AC RETURN AND S8)
AC PWR "OFF"	----	-----	---	ON	----------	---	------	
ACCESS MODE								
INTRUSION	OFF	OFF	ON	N/A	+24 to 28.5 V	OFF	20 ± 1	INTRUSION ALARMS INHIBITED
TAMPER	X[1]	X[3](X[1])[OFF]	ON	N/A	+24 to 28.5 V	X[3][OFF]	20 ± 1	
DURESS	X[1]	X[3](X[1])	ON	N/A	+24 to 28.5 V	OFF	20 ± 1	
NORMAL (NON-ALARM	OFF	OFF	ON	OFF	+24 to 28.5 V	OFF	20 ± 1	
AC PWR "ON"	----	-----	---	OFF	----------	---	------	(AFTER 3 SEC DELAY, RESET BY AC RETURN AND S8)
AC PWR "OFF"	----	-----	---	ON	----------	---	------	
TEST/RESET MODE								
INTRUSION	X[5]	X[1]	ON	N/A	0 V	OFF	20 ± 1	
TAMPER	X[5]	X[1]	ON	N/A	0 V	OFF	20 ± 1	
DURESS	X[5]	X[1]	ON	N/A	0 V	OFF	20 ± 1	
NORMAL (NON-ALARM	OFF	OFF	ON	OFF	0 V	OFF	20 ± 1	
AC PWR "ON"	----	-----	---	OFF	----------	---	------	(AFTER 3 SEC DELAY, RESET BY AC RETURN AND S8)
AC PWR "OFF"	----	-----	---	ON	----------	---	------	

NOTES

X INDICATES LED ON, ALARM CONDITION

-1 NO DELAY, ALARM APPEARS FOR 10 ± 2 SECONDS, CAN BE RESET BY S8 AFTER THAT TIME

-2 APPEARS AFTER 10-90 SECONDS DELAY, LATCHED UNTIL RESET BY TURNING MODE SWITCH TO TEST/RESET. CANNOT BE RESET BY S8 ALONE.

-3 NO DELAY, LATCHED UNTIL RESET BY TURNING MODE SWITCH TO TEST/RESET. CANNOT BE RESET BY S8 ALONE.

-4 APPEARS AFTER 10-90 SECONDS DELAY, REMAINS FOR 10 ± 2 SECONDS AFTER WHICH CAN BE RESET BY S8.

-5 SAME AS -1 EXCEPT INTERNAL ALARM ALSO SOUNDS FOR 10 ± 2 SECONDS.

( ) INDICATES LATCH/NON LATCH SWITCH IN NON LATCH POSITION.

[ ] INDICATES LAA INHIBIT SWITCH ON.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:10Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-C-52853B Control Unit, Alarm Set: C-9412/FSS-9(V)

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:10Z
Check the source to verify that this is the current version before use.